

CRI Bioinformatics Bootcamp 2026

Day 6: Introduction to Spatial Transcriptomics

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Creating a Seurat Object: QC and unsupervised clustering (Banksy and Sketching/Projecting)

Visium HD Spatial Gene Expression Library, Human Breast Cancer, IF (FFPE) - 10X genomics

<https://www.10xgenomics.com/datasets/visium-hd-cytassist-gene-expression-libraries-human-breast-cancer-ffpe-if>

Visium HD Spatial Gene Expression Library, Human Kidney (FFPE) - 10X genomics

<https://www.10xgenomics.com/datasets/visium-hd-cytassist-gene-expression-libraries-human-kidney-ffpe>

SpaceRanger Input: high resolution H&E (brightfield) or IF (Fluorescent) image ($\approx 1\text{GB}$) and FASTQ files (sometimes json files)

SpaceRanger Outputs: can be in customized bins, multiple of 2, up to $100\times 100\mu\text{m}$ bin

☐ Show Owner/Mode ☐ Show Dotfiles Filter:

Showing 3 rows - 0 rows selected

Type	Name	Size	Modified at
☐ 	square_016um	-	12/13/2024 2:02:00 AM
☐ 	square_008um	-	12/13/2024 2:01:51 AM
☐ 	square_002um	-	12/13/2024 2:01:26 AM

We choose the $8\mu\text{m}\times 8\mu\text{m}$ bin to achieve closer to a single cell and since it is the smallest bin that can produce a cLoupe Browser interactive file

Creating and QC-ing a Visium HD Seurat Object

Visium_HD_Human_Breast_Cancer_FFPE - Gene expression library of FFPE Human Breast Cancer (Visium HD) using the Human Whole Transcriptome Probe Set

Summary Image Alignment Bin-Level Metrics

Key Metrics

663,857

Number of 8 μ m binned squares under tissue

557.3

Mean reads per 8 μ m bin

96.9

Mean UMIs per 8 μ m bin

18,010

Total genes detected

Mapping [?]

Reads Mapped to Probe Set	97.3%
Reads Mapped Confidently to Probe Set	87.0%
Reads Mapped Confidently to the Filtered Probe Set	84.9%
Reads Half-Mapped to Probe Set	1.0%
Reads Split-Mapped to Probe Set	9.3%

Sequencing [?]

Number of Reads	369,943,706
Valid Barcodes	90.5%
Valid UMIs	99.9%
Sequencing Saturation	77.3%
Q30 Bases in Barcode	91.2%
Q30 Bases in Probe Read	89.7%
Q30 Bases in UMI	90.5%
Fraction of Bins Under Tissue 8 μ m	94.5%
UMIs per sq mm of Tissue	1,514,525
Reads per sq mm of Tissue	7,837,798
Fraction Reads in Squares Under Tissue	99.3%

Total UMI Count to Image Alignment

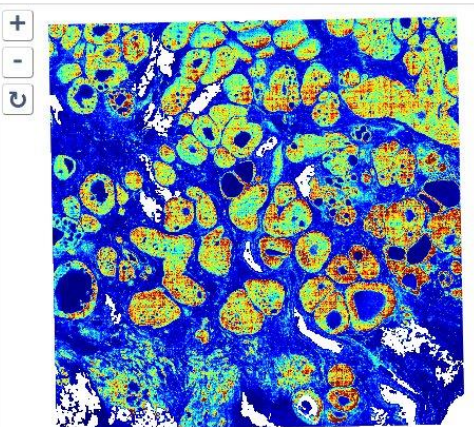
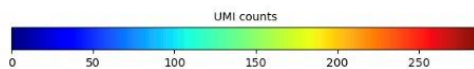
The total UMI count in each 8 μ m bin is overlaid onto the tissue image below to assess bin alignment and tissue detection. The highest value on the color scale corresponds to the 98th percentile of UMIs per 8 μ m bin under tissue, excluding bins with no UMIs.

Check that the overlay matches tissue morphology and covers all of the tissue of interest within the capture area.

If the overlay is inaccurate, use the manual fiducial alignment and tissue detection workflow in [Loupe Browser](#).

Additional QC images are in the Image Alignment tab for review.

All bins Bins under tissue UMI Color scale: Jet



Tissue Image 8 μ m bin UMI Counts

Visium_HD_Human_Breast_Cancer_FFPE - Gene expression library of FFPE Human Breast Cancer (Visium HD) using the Human Whole Transcriptome Probe Set

Summary Image Alignment Bin-Level Metrics

Bin Metrics Overview

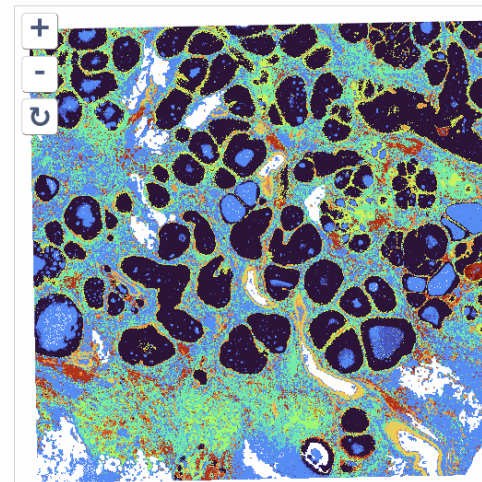
Bin Size (μ m)	8 μ m	16 μ m
Number of Bins Under Tissue	663,857	167,545
Mean UMI Counts per Bin	96.9	384.6
Mean Genes per Bin	89.9	327.0

Clustering

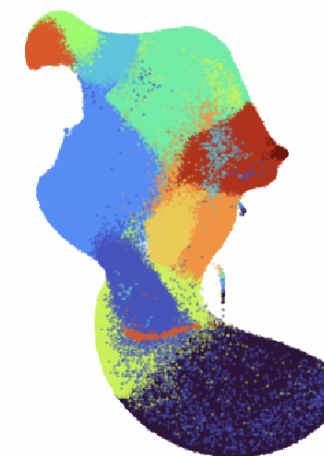
8 μ m bin

16 μ m bin

Tissue plot with 8 μ m bins colored by Graph-based clustering



UMAP Projection of 8 μ m bins colored by Graph-based clustering



Select All

- Cluster 1
- Cluster 2
- Cluster 3
- Cluster 4
- Cluster 5
- Cluster 6
- Cluster 7
- Cluster 8
- Cluster 9
- Cluster 10
- Cluster 11
- Cluster 12

Bin opacity: Display Tissue

New versions of SpaceRanger (v4.0 and higher) include **Stardist** segmentation within the outputs

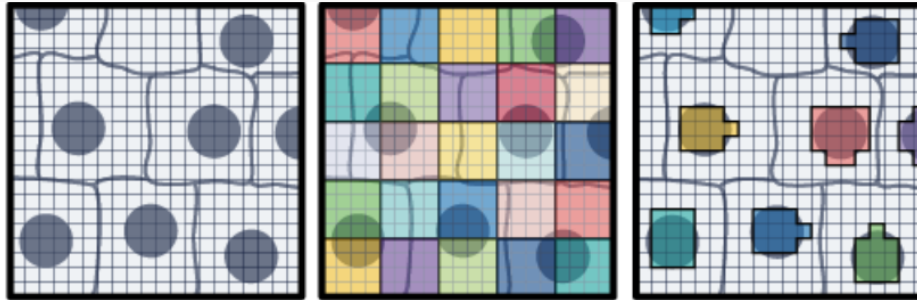
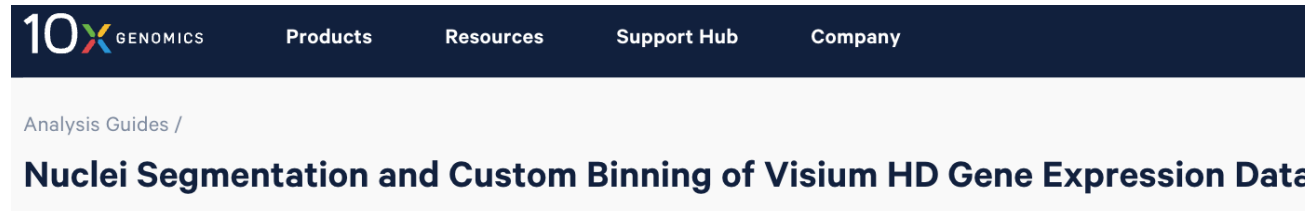


Figure 1: Two approaches for binning the $2 \times 2 \mu\text{m}$ barcode squares in Visium HD data. The capture area of a Visium HD slide is a continuous grid of barcodes (left panel). The Space Ranger pipeline by default will create square bins (shown here as $8 \times 8 \mu\text{m}$ squares of a single color) that tile the entire tissue containing capture area (middle panel). An alternative is to use the nuclei stain from a high-resolution H&E microscope image to group together barcodes that underlie the same cell nuclei (right panel).

Requirements

- High-resolution H&E image of the sample used in the Visium HD Gene Expression assay
- Output from a completed Space Ranger count analysis
- Familiarity with Python programming

The image taken by the CytAssist is not suitable for this analysis, because it is only a cytoplasmic eosin stain and lacks the hematoxylin stain for cell nuclei.

New versions of SpaceRanger (v4.0 and higher) include **Stardist** segmentation within the outputs

Summary Image Alignment Bin-Level Metrics **Cell Segmentation**

Cell Segmentation Metrics ?

Number of Cells	95,176
Reads in Cells	70.0%
UMIs in Cells	82.4%
Mean Reads per Cell	2,795.1
Median Genes per Cell	173.0
Median UMIs per Cell	212.0
Median Cell Area (μm^2)	56.0
Median Nucleus Area (μm^2)	12.0
Maximum Nucleus Diameter (pixels)	256

Cell Segmentation ?

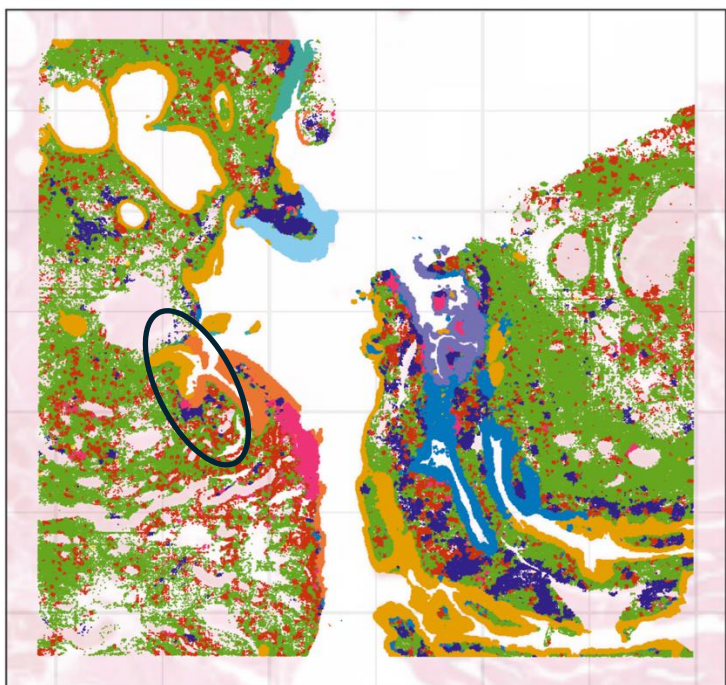


	Type	Name
☐	Folder	binned_outputs
☐	Folder	segmented_outputs
☐	Folder	spatial
☐	File	barcode_mappings.parquet
☐	File	cloupe_008um.cloupe
☐	File	cloupe_cell.cloupe
☐	File	feature_slice.h5
☐	File	metrics_summary.csv
☐	File	molecule_info.h5

However, Stardist might not be the ideal tool for the tissue of choice. Always check the tissue type and optimize cell segmentation accordingly

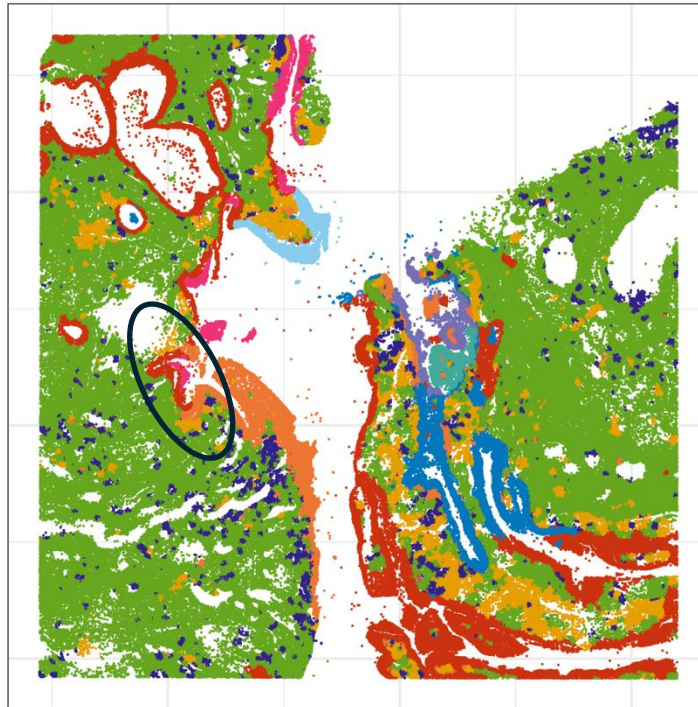
For the segmentation part in SpaceRanger, a json file from any segmentation tool can be provided and will yield a “segmented outs” folder for downstream analysis with third party tools

8μm bin



Banksy Cluster

- 0
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10



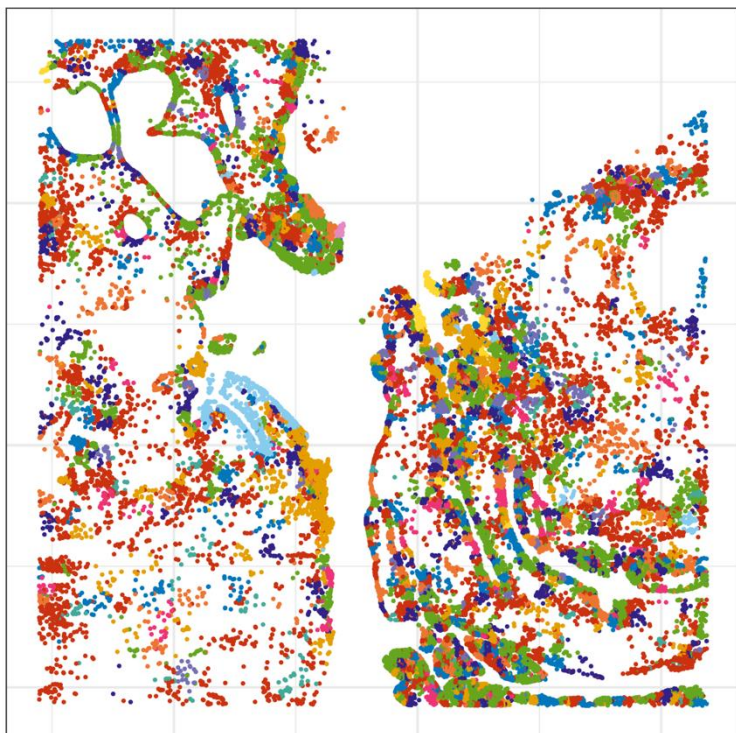
Banksy

- 0
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9

Qupath

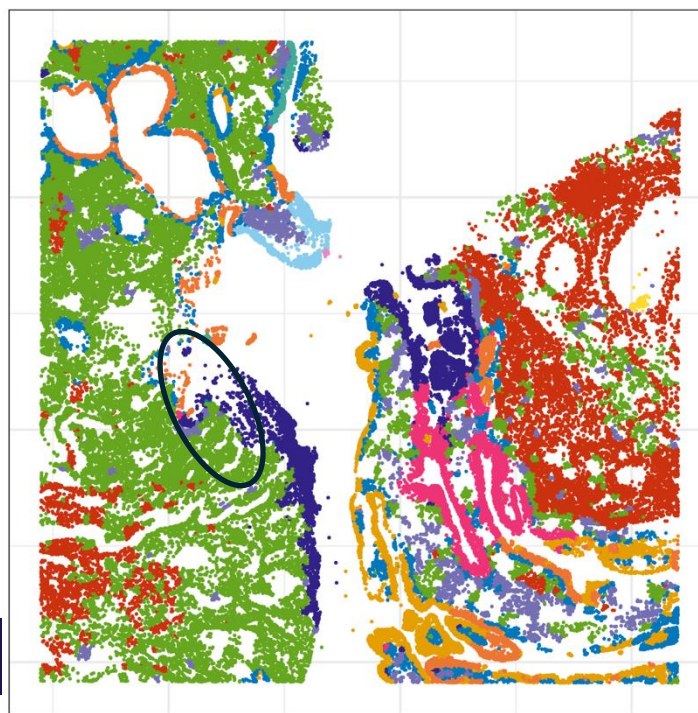
Banksy
Comparision

Instaseg



Banksy

- 0
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11



Banksy

- 0
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11

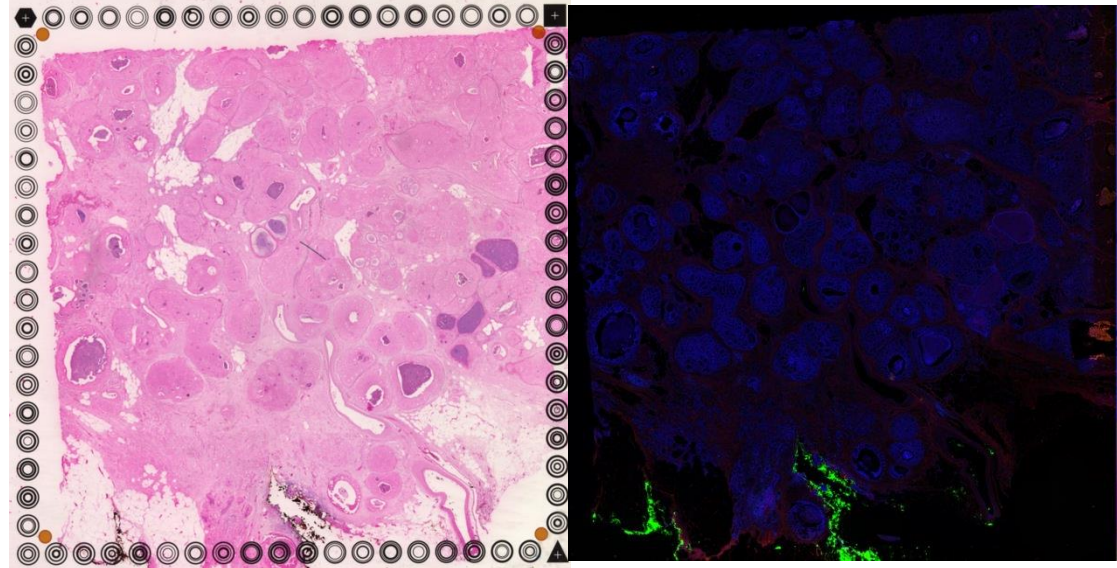
10X Stardist

Interactive Session

```
#####
##### Visium HD Seurat #####
#####
library(Seurat)
```

```
dirs <- c("/data1/shahs3/users/naseran1/work/projects/CRI_Spatial/data/Visium_HD_Breast_dataset_CRI/Output_Files/")
```

```
# Load individual sample
object <- Load10X_Spatial(data.dir = dirs, bin.size = c(8))
```

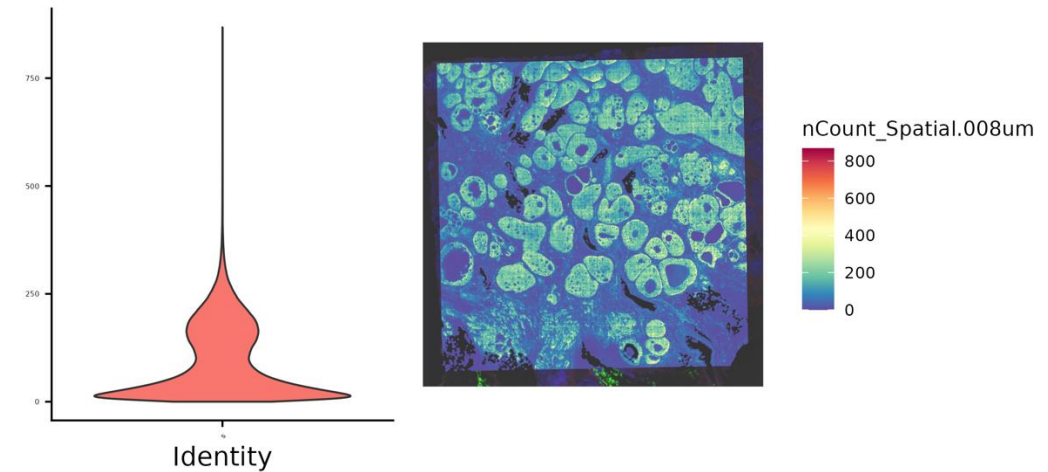


```
##### View the raw counts and features/genes
```

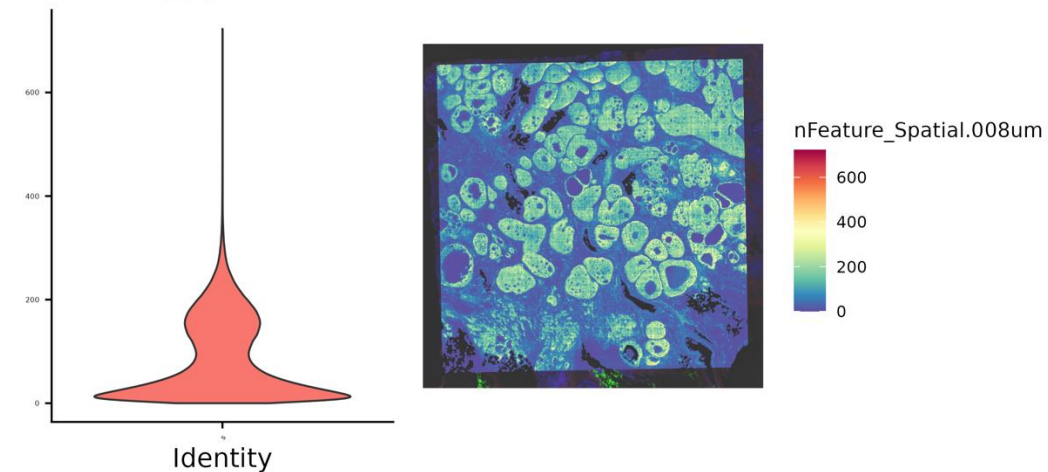
```
vln.plot1 <- VlnPlot(object, features = "nCount_Spatial.008um", pt.size = 0, raster=FALSE)
count.plot <- SpatialFeaturePlot(object, features = "nCount_Spatial.008um", pt.size.factor = 4,
                                image.alpha = 0.8, alpha=c(1,1))
vln.plot2 <- VlnPlot(object, features = "nFeature_Spatial.008um", pt.size = 0, raster=FALSE)
feature.plot <- SpatialFeaturePlot(object, features = "nFeature_Spatial.008um", pt.size.factor = 4,
                                image.alpha = 0.8, alpha=c(1,1))
```

```
(vln.plot1 | count.plot) / (vln.plot2 | feature.plot)
```

nCount_Spatial.008um



nFeature_Spatial.008um



Seurat Object QC

```
#####  
#####      Object QC      #####  
#####
```

```
# Add % MT
```

```
object[["MT.percent"]] <- PercentageFeatureSet(object, pattern = "^MT-")
```

```
# UMI and Gene Threshold
```

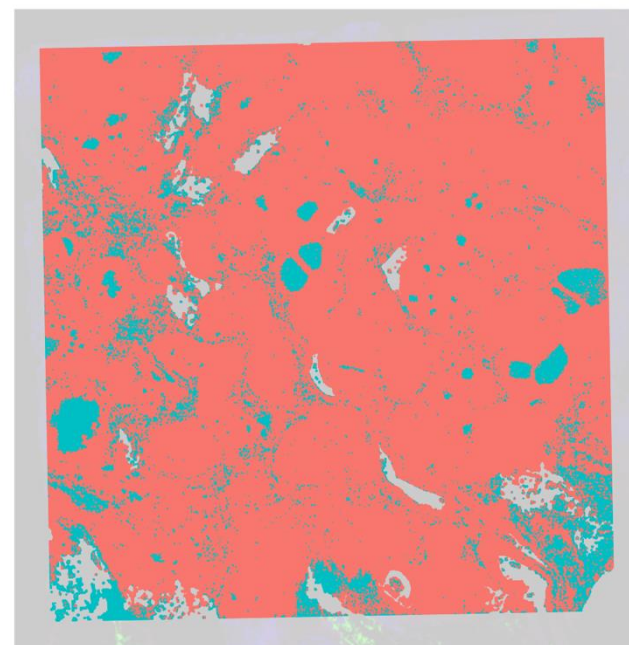
```
object$QCFilter<-ifelse(object$MT.percent < 25 &  
                        object$nCount_Spatial.008um > 10 &  
                        object$nFeature_Spatial.008um > 10 , "Keep", "Remove")
```

```
cells_plot <- object[[]] %>% filter(QCFilter == "Keep") %>% rownames(.)
```

```
p1 <- SpatialDimPlot(object, group.by = "QCFilter", pt.size.factor = 4, image.alpha = 0.2, alpha=c(1,0.2)) +  
  theme(  
    legend.position = "right",  
    legend.key.size = unit(0.3, "cm"),  
    plot.title = element_text(size = 16, hjust = 0.5, family = "Arial"),  
    legend.text = element_text(size = 14, family = "Arial"),  
    axis.text = element_text(size = 14, family = "Arial"),  
    legend.title = element_text(size = 14, family = "Arial"),  
    axis.title = element_text(size = 14, family = "Arial")) +  
  guides(fill = guide_legend(override.aes = list(size = 4)))
```

```
ggsave(filename = file.path(sample_dir, "QC.png"), plot = p1, width = 5, height = 5, dpi = 300)
```

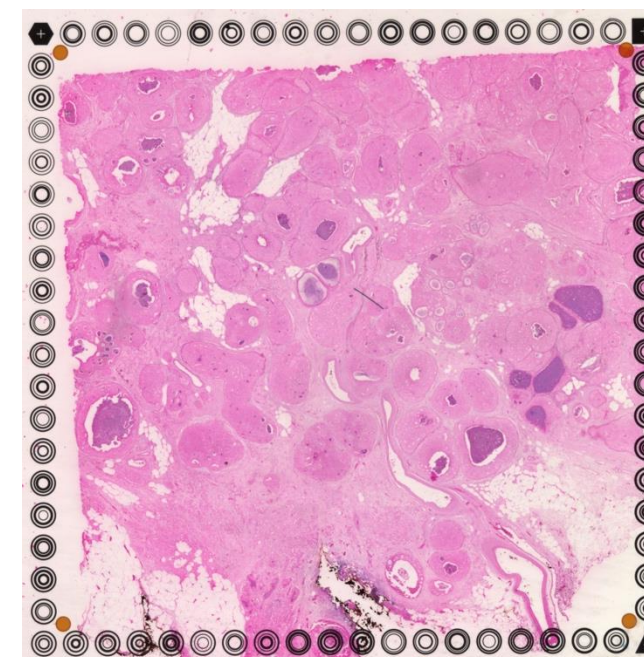
```
object<-subset(object,cells=colnames(object)[object$QCFilter=="Keep"])  
object <- UpdateSeuratObject(object)
```



QCFilter

● Keep

● Remove



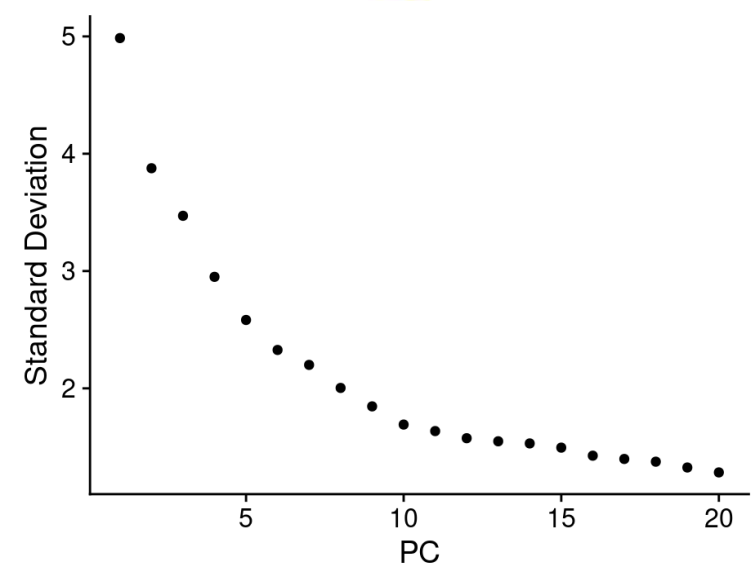
Banksy unsupervised neighborhood clustering (single object)

```
##### Banksy Clustering #####
#####
```

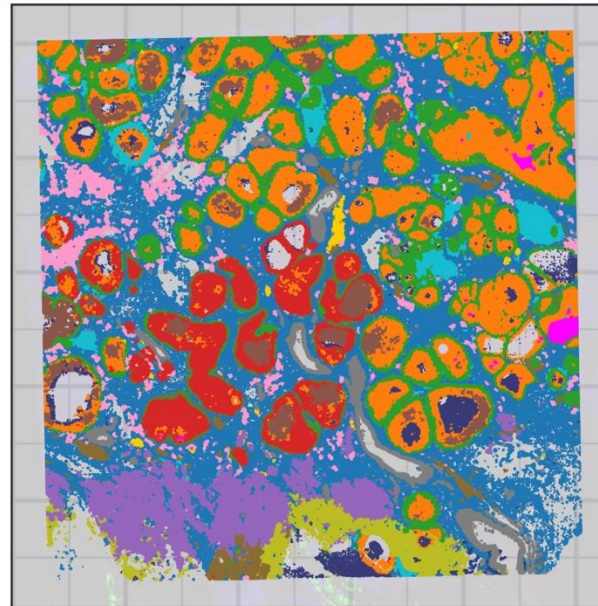
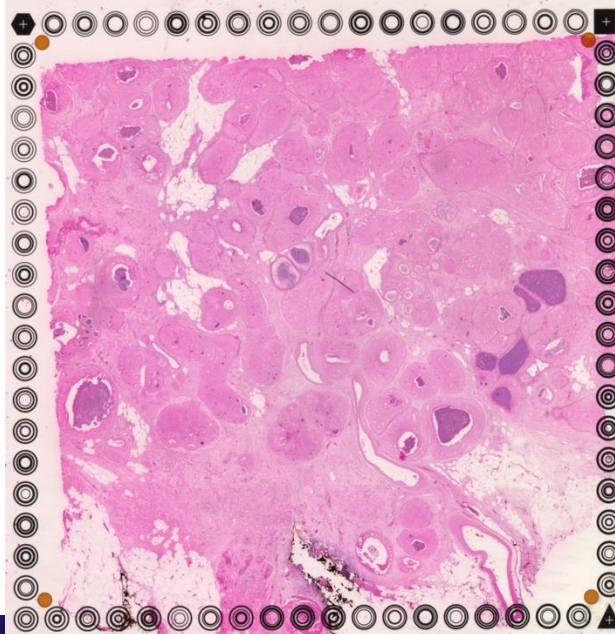
```
DefaultAssay(object) <- "Spatial.008um"
object <- NormalizeData(object)
object <- FindVariableFeatures(object)
```

```
# Run Banksy
object <- RunBanksy(object,
  lambda = 0.8, verbose = TRUE,
  assay = "Spatial.008um", slot = "data", features = "variable",
  k_geom = 24)
```

```
DefaultAssay(object) <- "BANKSY"
object <- RunPCA(object, assay = "BANKSY", reduction.name = "pca.banksy", features = rownames(object), npcs = 20)
ElbowPlot(object, reduction = "pca.banksy")
object <- FindNeighbors(object, reduction = "pca.banksy", dims = 1:15)
object <- FindClusters(object, cluster.name = "banksy_cluster", resolution = 0.3)
object <- RunUMAP(object, reduction = "pca.banksy", reduction.name = "umap.banksy", return.model = T, dims = 1:15)
```



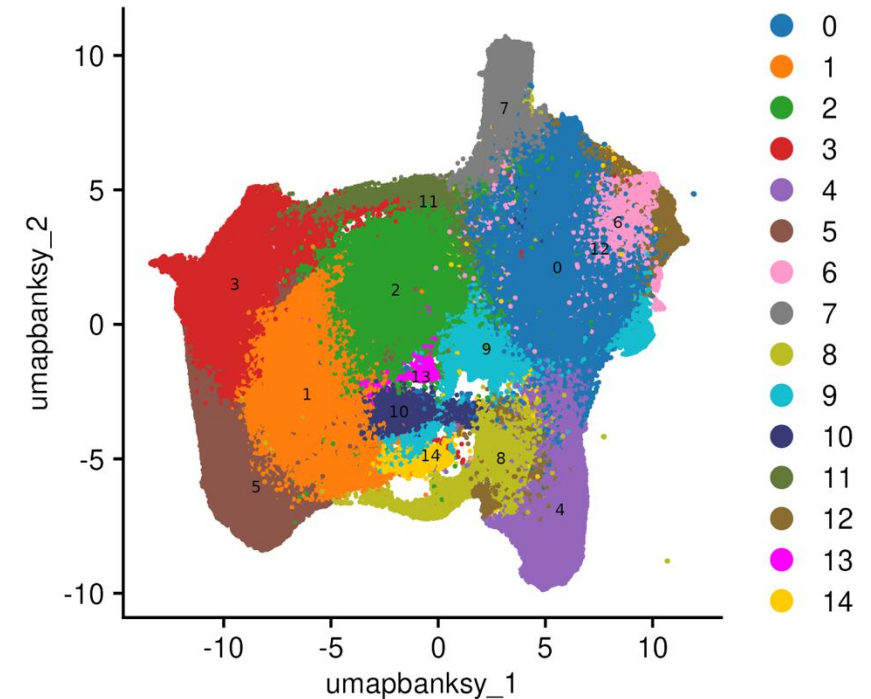
Banksy Neighborhood Clustering



banksy_cluster



Banksy Neighborhood Clustering



Seurat Unsupervised Clustering: Sketching and Projection (single object)

object	S4 [4000 x 593942] (SeuratO	S4 object of class Seurat
assays	list [2]	List of length 2
Spatial.008um	S4 [18085 x 593942] (SeuratO	S4 object of class Assay5
BANKSY	S4 [4000 x 593942] (SeuratO	S4 object of class Assay
meta.data	list [593942 x 8] (S3: data.fra	A data.frame with 593942 rows and 8 columns
active.assay	character [1]	'BANKSY'
active.ident	factor	Factor with 15 levels: "0", "1", "2", "3", "4", "5", ...
graphs	list [2]	List of length 2
neighbors	list [0]	List of length 0
reductions	list [2]	List of length 2
images	list [1]	List of length 1
project.name	character [1]	'SeuratProject'
misc	list [0]	List of length 0
version	list [1] (S3: package_version, i	List of length 1
commands	list [6]	List of length 6
tools	list [0]	List of length 0

```
#####
##### Sketching followed by Projection #####
#####
```

```
# note that data is already normalized but not scaled
```

```
DefaultAssay(object) <- "Spatial.008um"
```

```
object <- FindVariableFeatures(object)
```

```
object <- ScaleData(object)
```

```
# we select 100,000 cells and create a new 'sketch' assay
```

```
object <- SketchData(
  object = object,
  ncells = 100000,
  method = "LeverageScore",
  sketched.assay = "sketch"
)
```

```
# switch analysis to sketched cells
```

```
DefaultAssay(object) <- "sketch"
```

```
# perform clustering workflow
```

```
object <- FindVariableFeatures(object)
```

```
object <- ScaleData(object)
```

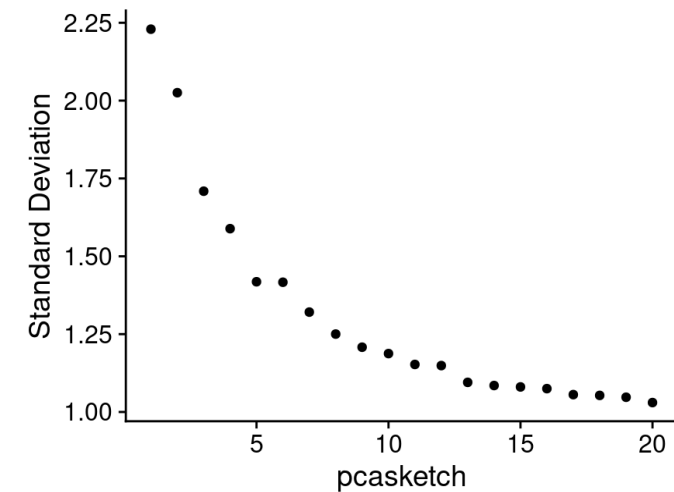
```
object <- RunPCA(object, assay = "sketch", reduction.name = "pca.sketch")
```

```
ElbowPlot(object, reduction = "pca.sketch")
```

```
object <- FindNeighbors(object, assay = "sketch", reduction = "pca.sketch", dims = 1:15)
```

```
object <- FindClusters(object, cluster.name = "seurat_cluster.sketched", resolution = 0.2)
```

```
object <- RunUMAP(object, reduction = "pca.sketch", reduction.name = "umap.sketch", return.model = T, dims = 1:15)
```

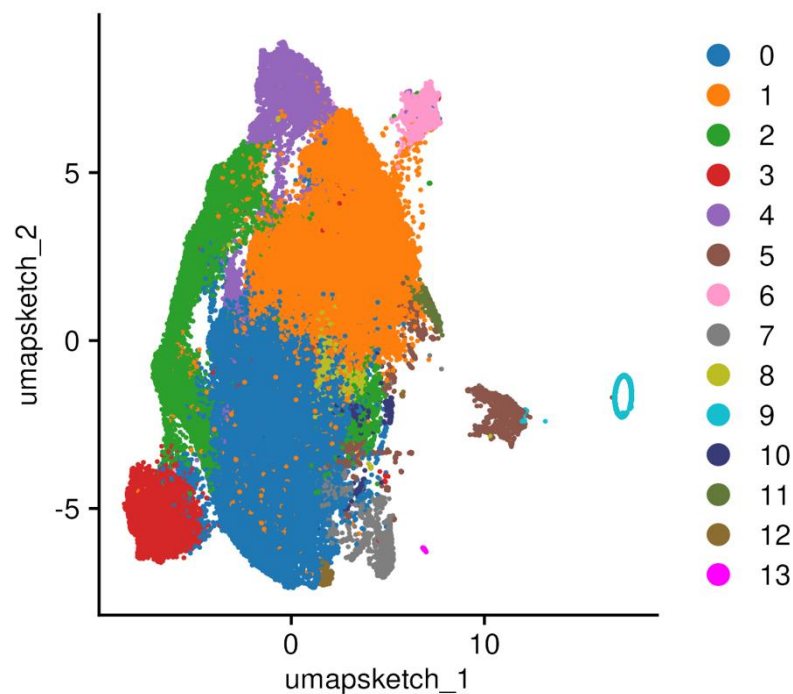


```
options(future.globals.maxSize = 3 * 1024^3)
```

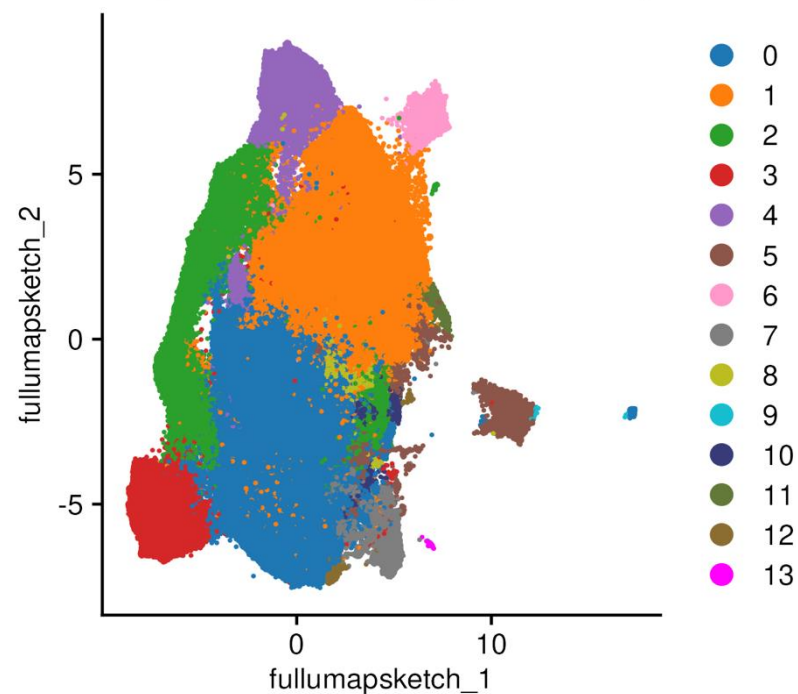
```
object <- ProjectData(  
  object = object,  
  assay = "Spatial.008um",  
  full.reduction = "full.pca.sketch",  
  sketched.assay = "sketch",  
  sketched.reduction = "pca.sketch",  
  umap.model = "umap.sketch",  
  dims = 1:15,  
  refdata = list(seurat_cluster.projected = "seurat_cluster.sketched")  
)
```

object	S4 [18085 x 593942] (SeuratO	S4 object of class Seurat
assays	list [3]	List of length 3
Spatial.008um	S4 [18085 x 593942] (SeuratO	S4 object of class Assay5
BANKSY	S4 [4000 x 593942] (SeuratO	S4 object of class Assay
sketch	S4 [18085 x 1e+05] (SeuratO	S4 object of class Assay5
object	Large Seurat	(53.5 GB)

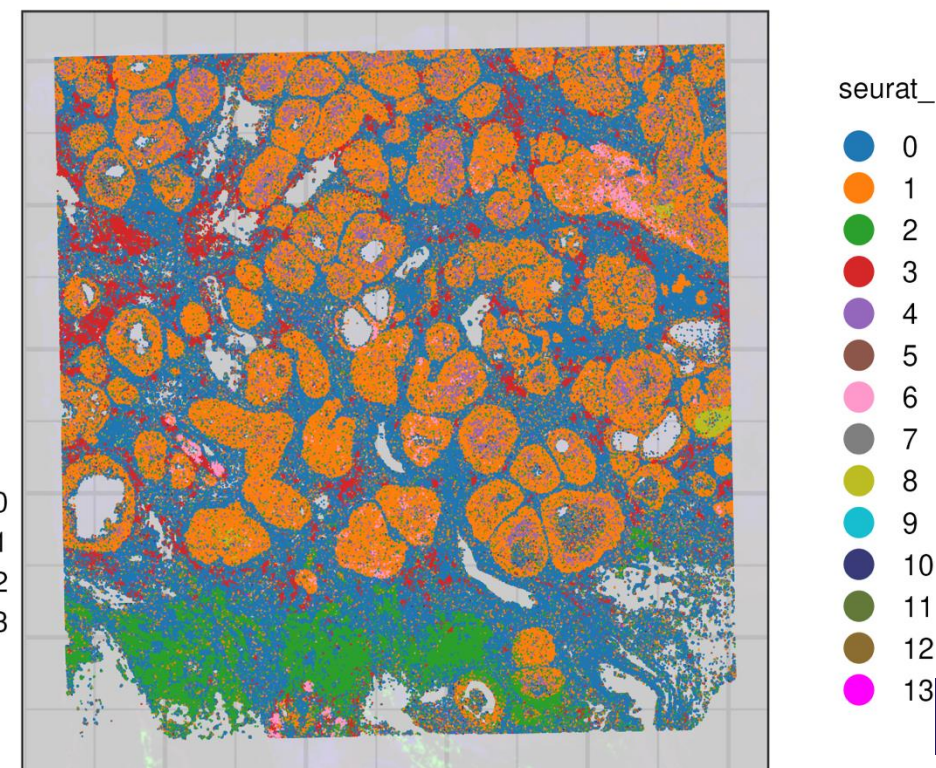
Sketched clustering (100,000 cells)



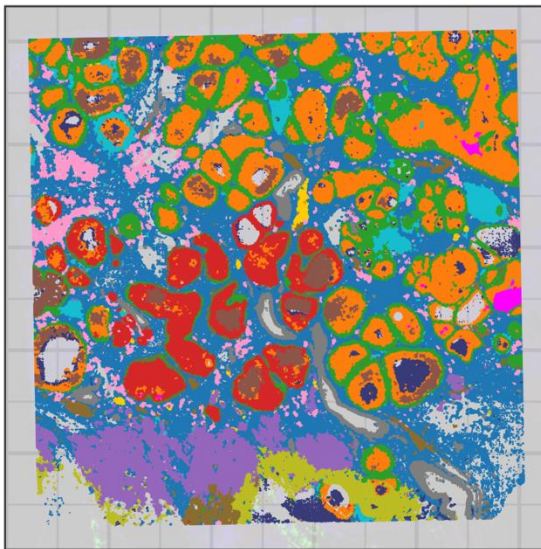
Projected clustering (full dataset)



Projected clustering (full dataset)



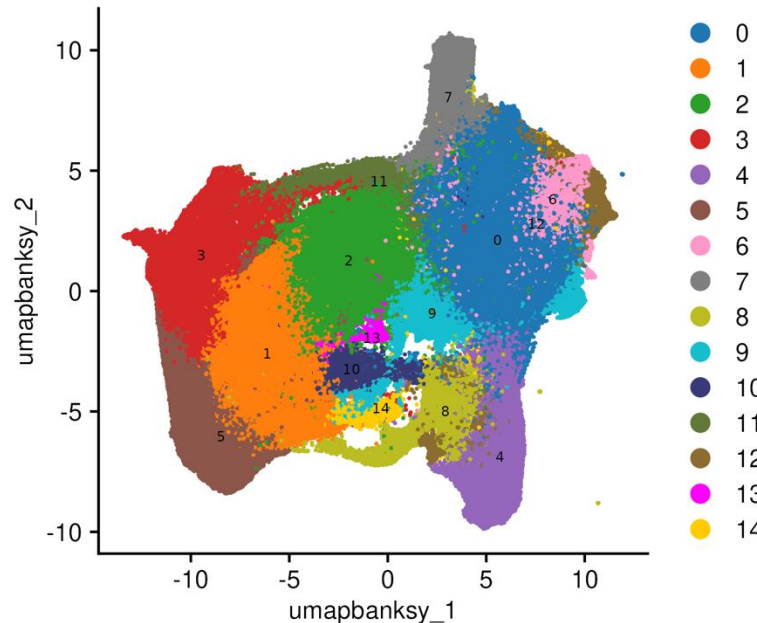
Banksy Neighborhood Clustering



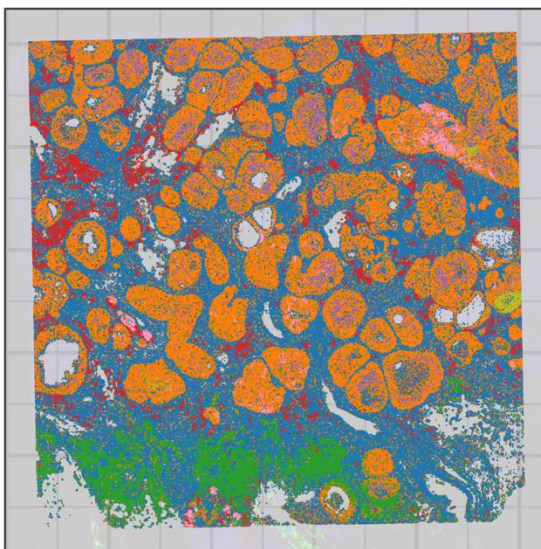
banksy_cluster



Banksy Neighborhood Clustering



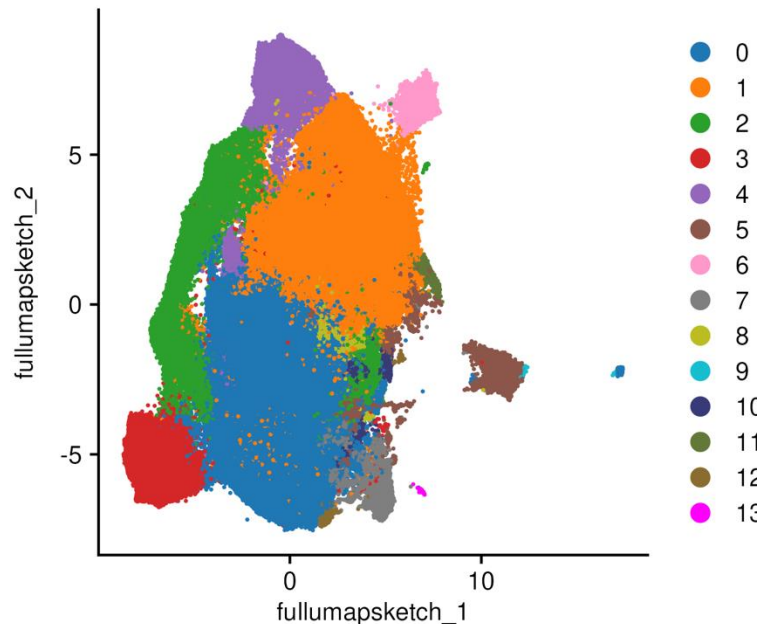
Projected clustering (full dataset)



seurat_cluster.projected



Projected clustering (full dataset)



Banksy vs Sketching and Projection:

Banksy unsupervised clustering takes into account neighborhood, shared transcriptional signatures, and rare cell types

Seurat's built-in unsupervised clustering only takes into account shared transcriptional signatures

Cell Typing, CNV, and Trajectory Analysis

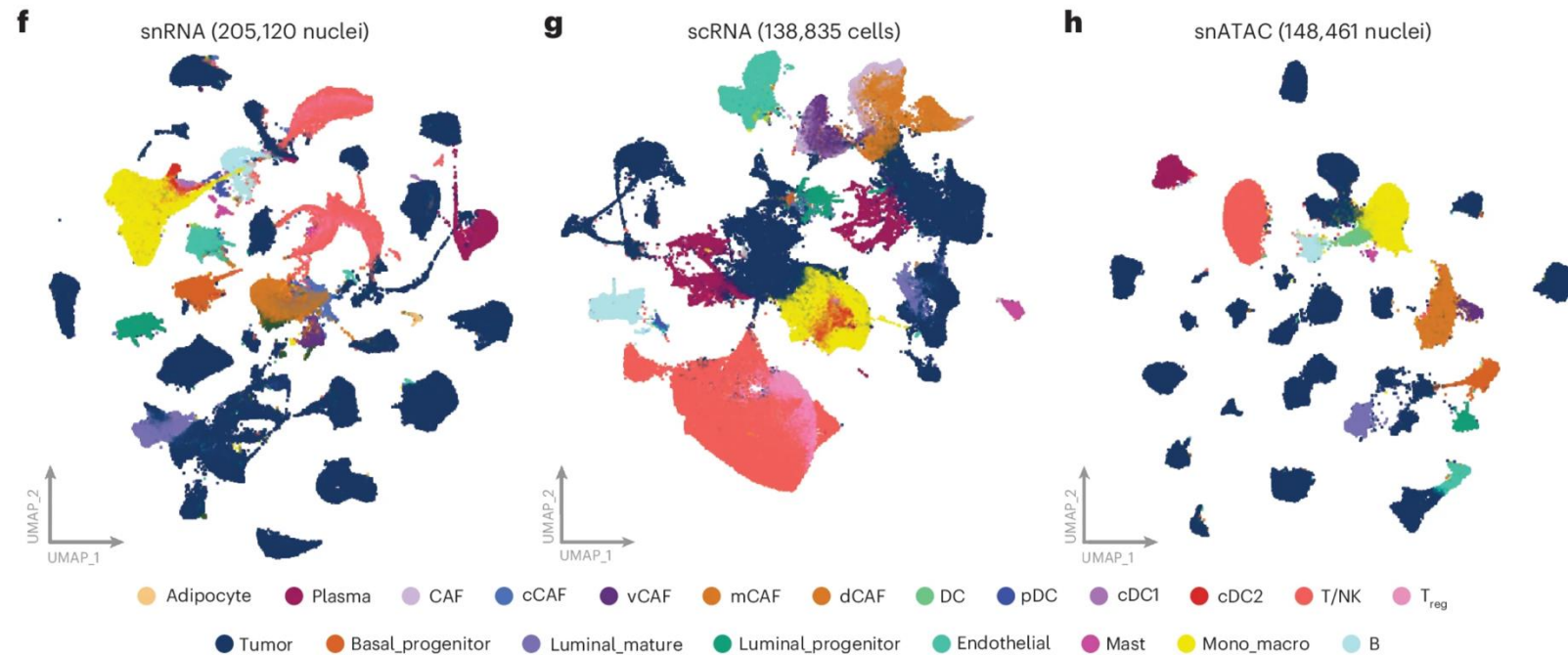
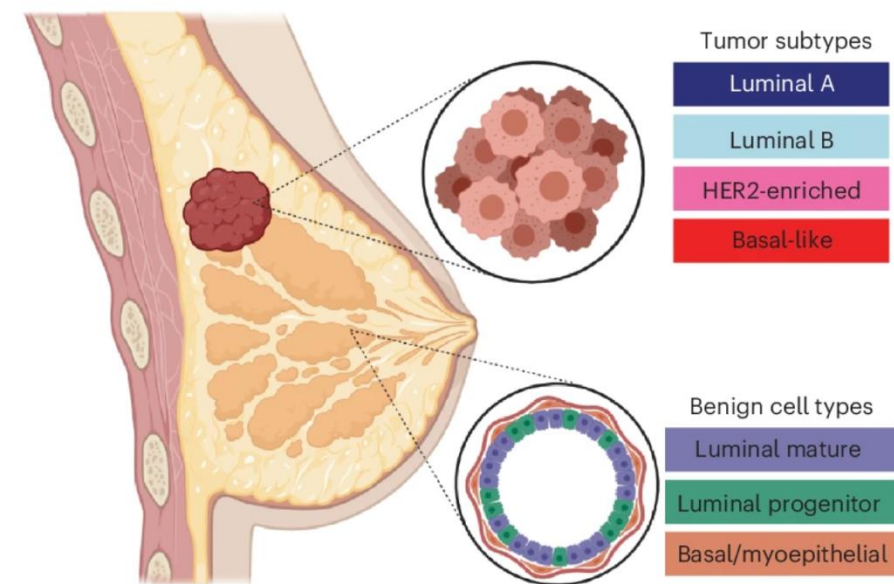
Gene-set Supervised Cell Type Annotation

Differential chromatin accessibility and transcriptional dynamics define breast cancer subtypes and their lineages

[Michael D. Iglesia](#), [Reyka G. Jayasinghe](#), [Siqi Chen](#), [Nadezhda V. Terekhanova](#), [John M. Herndon](#), [Erik Storrs](#), [Alla Karpova](#), [Daniel Cui Zhou](#), [Nataly Naser Al Deen](#), [Andrew T. Shinkle](#), [Rita Jui-Hsien Lu](#), [Wagma Caravan](#), [Andrew Houston](#), [Yanyan Zhao](#), [Kazuhito Sato](#), [Preet Lal](#), [Cherease Street](#), [Fernanda Martins Rodrigues](#), [Austin N. Southard-Smith](#), [André Luiz N. Targino da Costa](#), [Houxiang Zhu](#), [Chia-Kuei Mo](#), [Lisa Crowson](#), [Robert S. Fulton](#), ... [Li Ding](#) 

[+ Show authors](#)

[Nature Cancer](#) **5**, 1713–1736 (2024) | [Cite this article](#)



Step3: Supervised cell type assignment using well defined gene sets with AddModuleScore

```
## Breast cancer gene set from Iglesia MD ... Naser Al Deen N et al. Nature Cancer. 2024 Nov;5(11):1713-36.

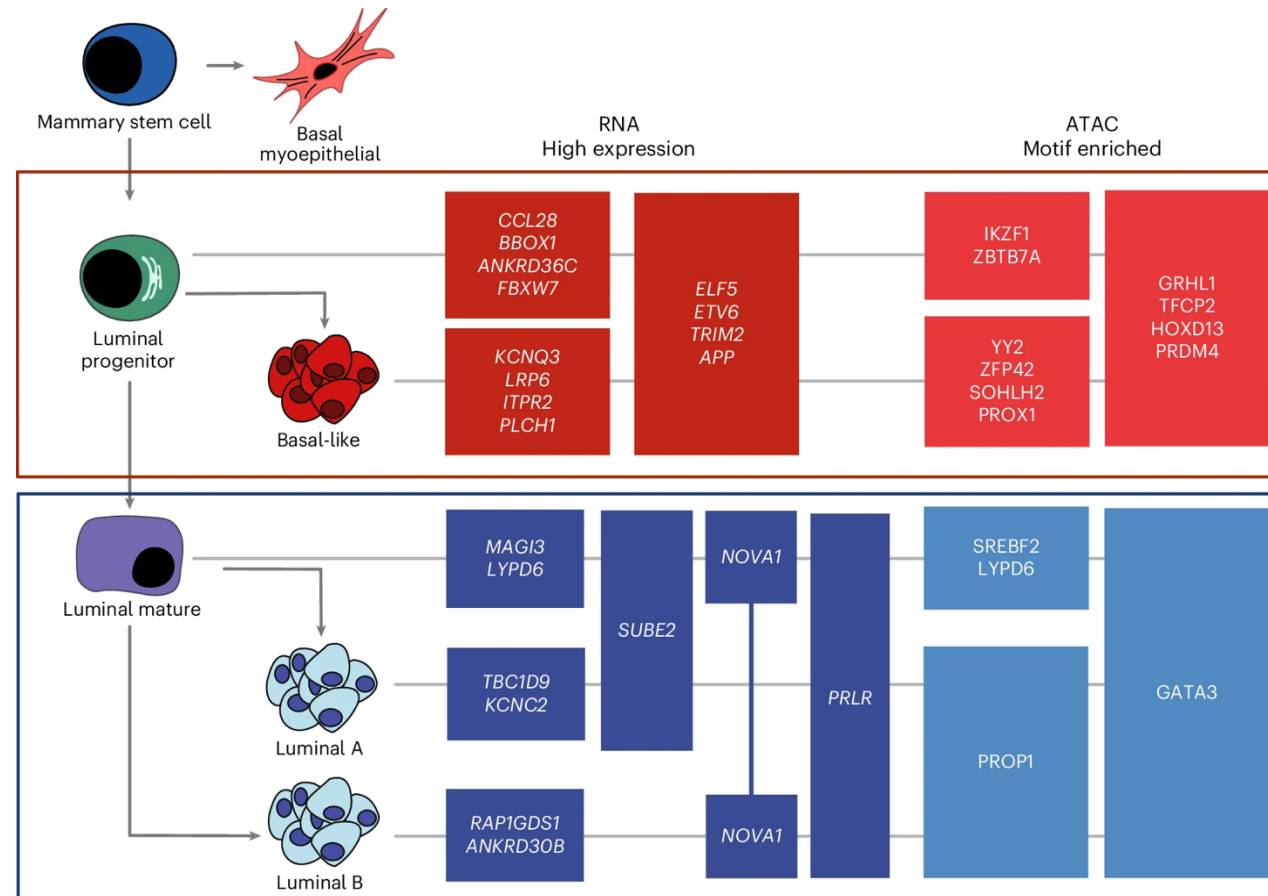
#####
##### DCIS Major Cell types #####
#####

library(writexl)
library(openxlsx)
library(readxl)

gene_lists <- list(
  CD8.T.cells = c("CD8B", "CD8A", "CD3E", "CD3D"),
  CD4.T.cells = c("CD4", "CD3E", "CD3D", "SELL", "CCR7", "IL7R", "TCF7", "LEF1"),
  NK.cells = c("XCL2", "XCL1", "KLRF1", "KIR2DL3", "IL2RB", "CD7", "KLRB1", "KLRD1",
    "GZMA", "PRF1", "CD160", "NCAM1"),
  B.cell = c("BANK1", "CD79A", "CD74", "MS4A1", "MEF2C", "CD19", "CD79B"),
  Plasma = c("IGKC", "IGLC3", "IGLC2", "IGHG1", "IGHA1", "IGHG3", "IGHG4", "IGHA2", "FCRL5", "TNFRSF17"),
  Mast.cells = c("TPSB2", "TPSAB1", "CPA3", "MS4A2", "HPGDS", "KIT", "ENPP3"),
  Macrophages = c("ITGAM", "LGALS3", "CD68", "CD163", "LYZ", "ADGRE1", "LAMP2"),
  Monocytes = c("CD14", "FCGR3A", "FCGR1A"),
  DCs = c("PTGDS", "FCHSD2", "GPR183", "NR3C1", "TCF4"),
  Endothelial.cells = c("FABP4", "VWF", "ACKR1", "LDB2", "PECAM1"),
  CAF = c("CFD", "DCN", "GSN", "EBF1", "PRKG1"),
  Basal_Myoepithelial = c("KRT14", "DST", "MMP7", "MIR205HG", "MT1X", "OXTR", "KRT17", "FST"),
  Epithelial.cells = c("EPCAM", "AMBP", "MUC1"),
  Fibroblasts = c("TIMP1", "FN1", "POSTN", "ACTA2", "BST2", "LY6D", "COL6A1", "SLC20A1", "COL6A2", "KRT16", "CD9",
    "S100A4", "EMP1", "LRRC8A", "EPCAM", "PDPN", "ITGB1", "PDGFRA", "THY1"),
  Adipocytes = c("PNPLA2", "CAV1", "FABP4", "PPARG", "CEBPA", "LEP", "CIDEA", "SHOX2", "SLC7A10", "SLC36A2", "P2RX5"),
  Tumor = c("MYC", "ESR1", "AR", "PGR", "CDH1", "AKT1", "ERBB2", "EPCAM", "KRT8", "KRT18", "KRT19"),
  Luminal.progenitor = c("MGP", "SCGB2A2", "SLPI", "LTF", "PTN", "KIT", "ALDH1A3"),
  Luminal.mature = c("MUCL1", "MGP", "ERBB4", "ANKRD30A", "AZGP1", "AGR2", "STC2"))

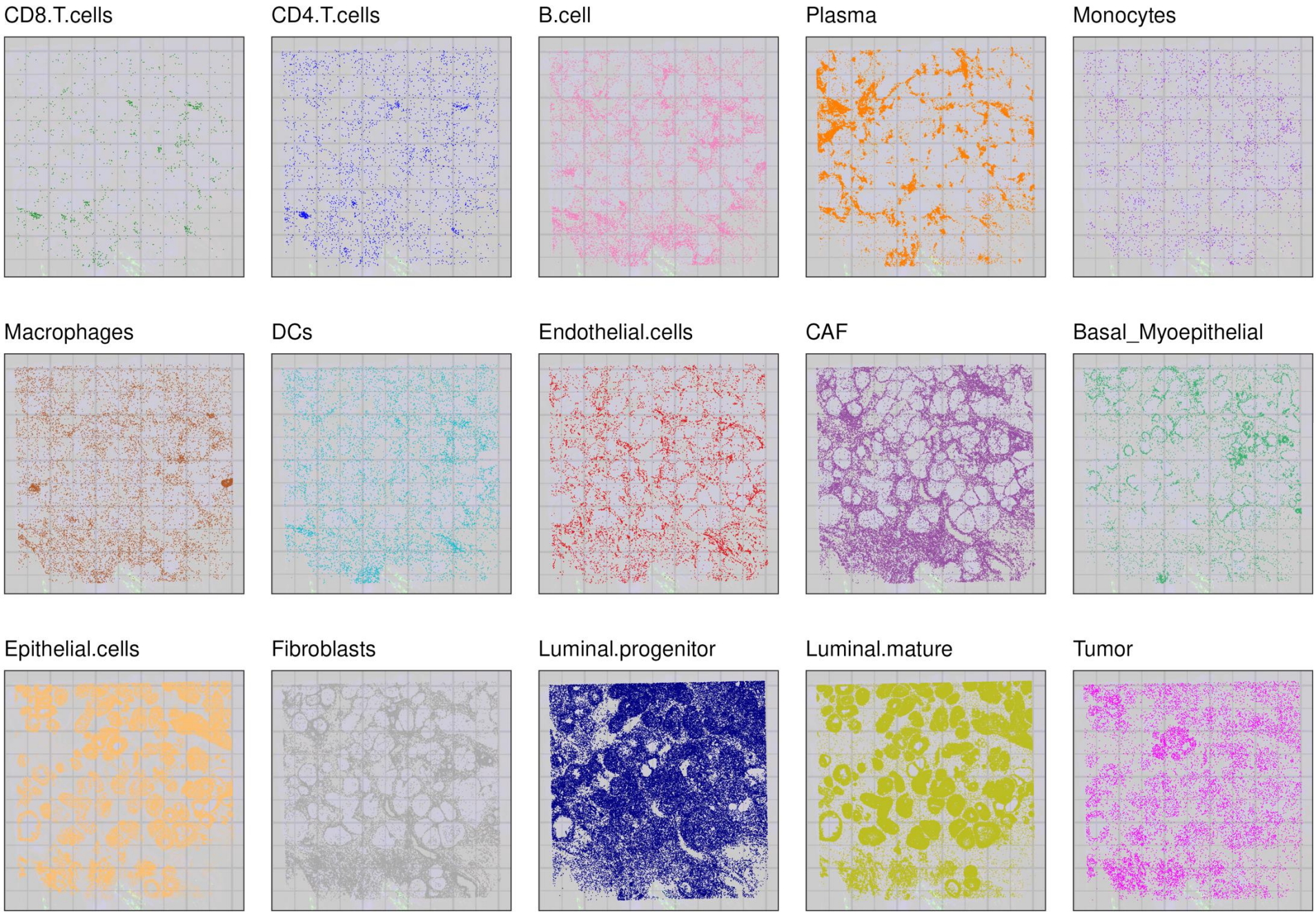
# Convert the list into a data frame
genes_df <- do.call(rbind, lapply(names(gene_lists), function(cell_type) {
  data.frame(Gene = gene_lists[[cell_type]], Cell.type = cell_type, stringsAsFactors = FALSE)
}))

# Write the data frame to an Excel file
write_xlsx(genes_df, "/data1/shahs3/users/naseran1/work/projects/CRI_Spatial/objects/DCIS_genes.xlsx")
```



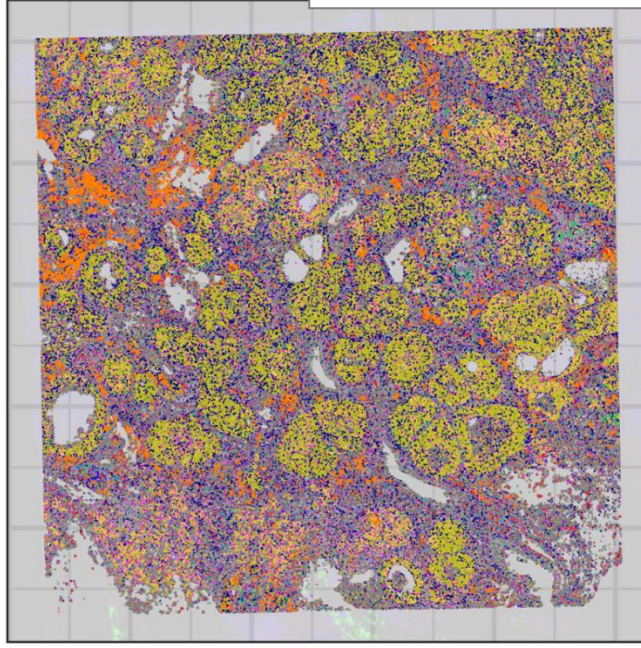
Iglesia MD et al. Nature Cancer. 2024 Nov;5(11):1713-36.

Supervised (gene-list informed) cell type annotation using AddModuleScore

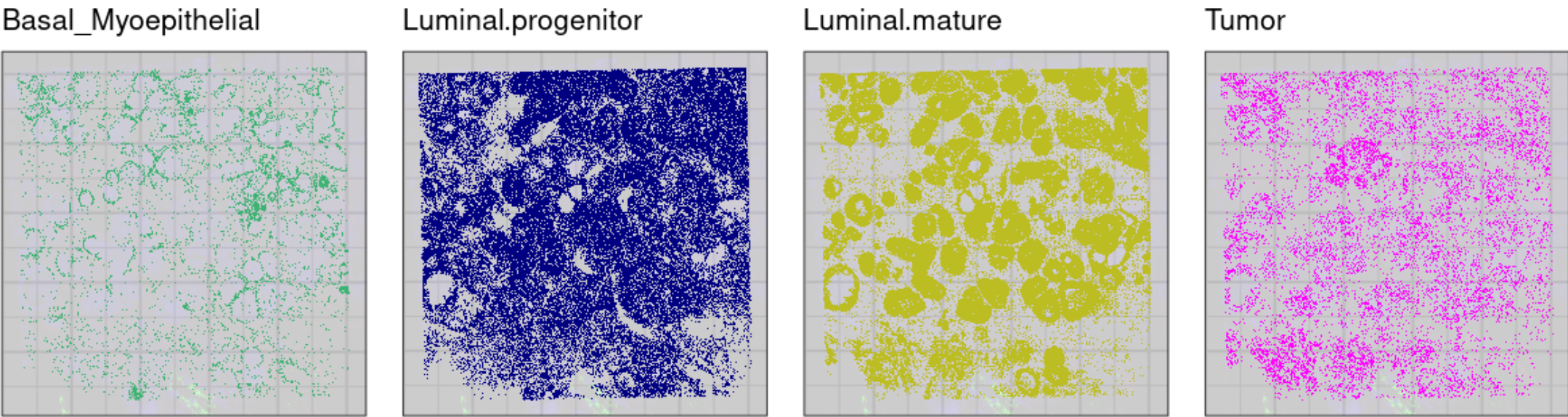


- cell_type
- CD8.T.cells
 - CD4.T.cells
 - B.cell
 - Plasma
 - Monocytes
 - Macrophages
 - DCs
 - Endothelial.cells
 - CAF
 - Basal_Myoepithelial
 - Epithelial.cells
 - Fibroblasts
 - Luminal.progenitor
 - Luminal.mature
 - Tumor
 - NA

Major Cell Types

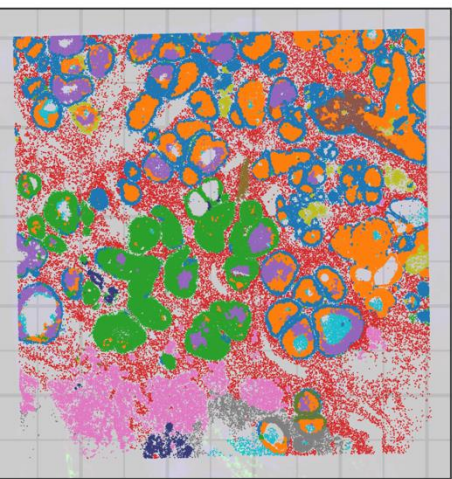


Subsetting into epithelial cells only and re-clustering based on Banksy neighborhood

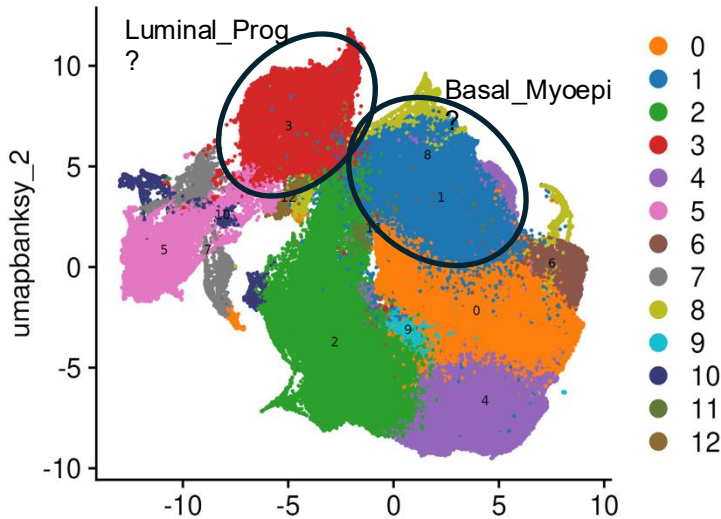


The rest is a mix of all cell types except for the Basal_Myoepithelial – Is cluster 5 IDC (doesn't seem to have an intact myoepithelium) - check DEGs and hallmark of cancer pathways

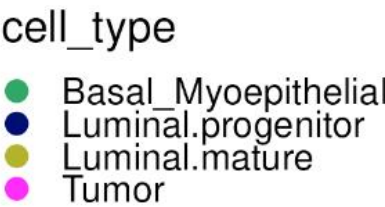
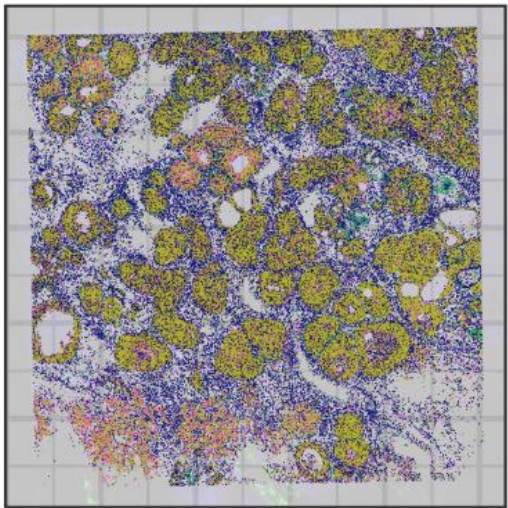
Banksy Neighborhood Clustering



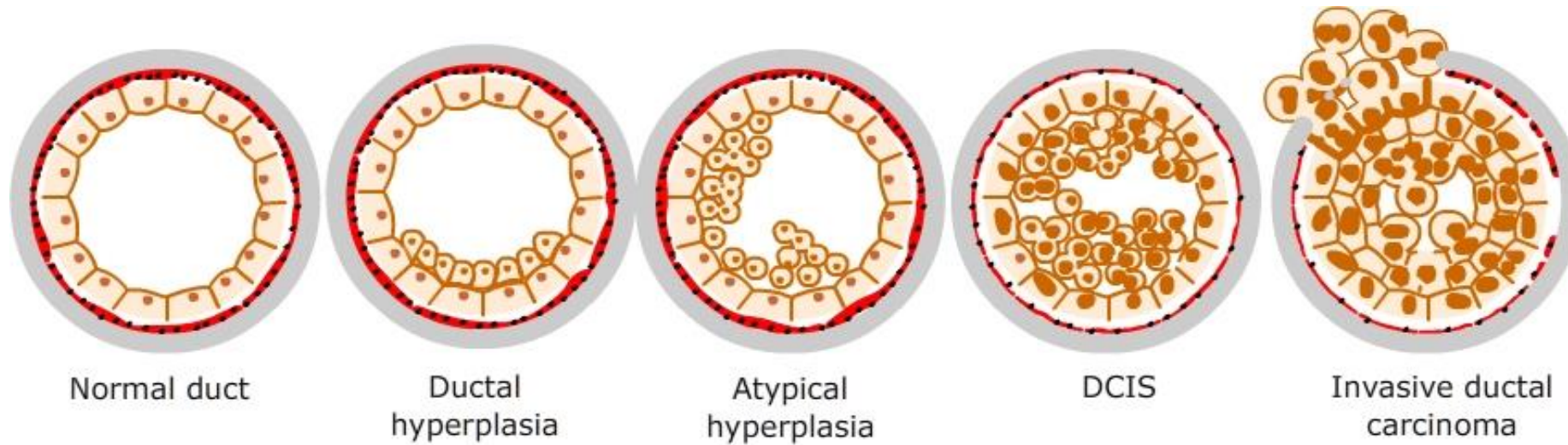
Banksy Neighborhood Clustering



Major Cell Types



Ductal carcinoma in situ progression



Basement membrane

Myoepithelium

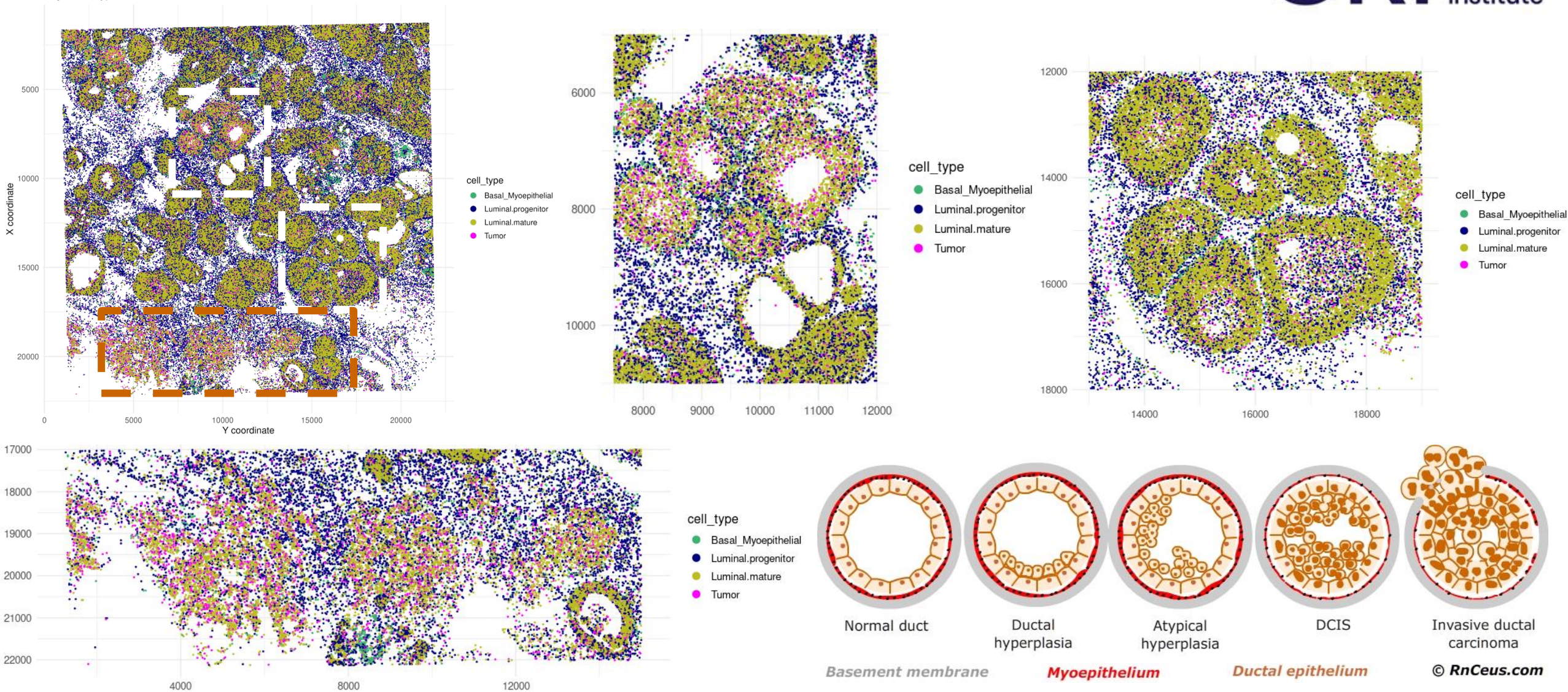
Ductal epithelium

© **RnCeus.com**

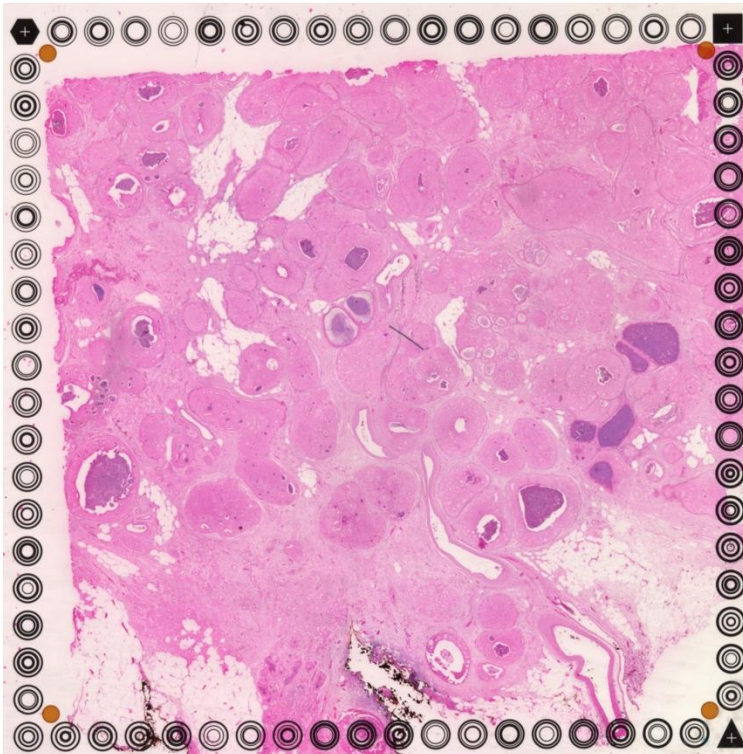
Zooming into certain regions of interest (ROIs)

`coordinates <- GetTissueCoordinates(object)`

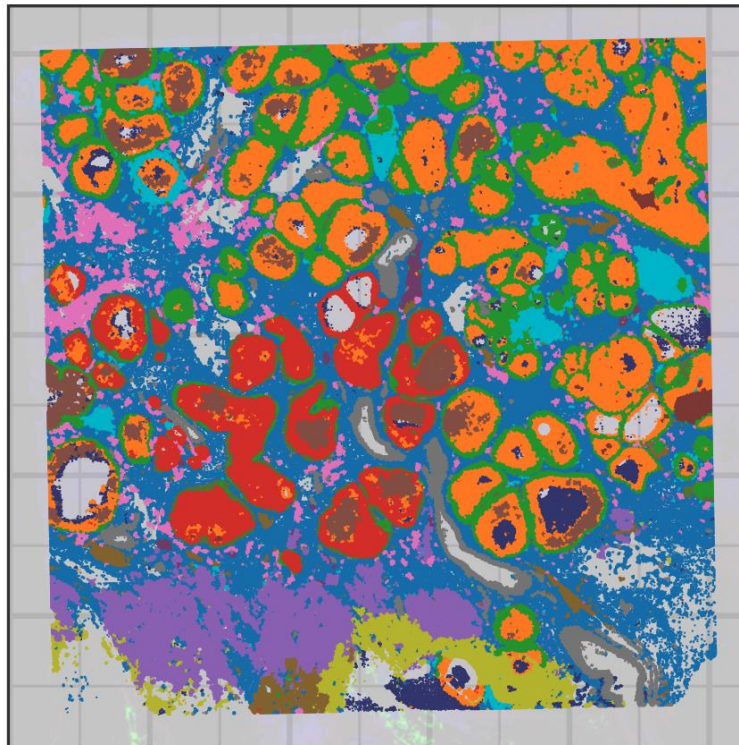
Major Cell Types



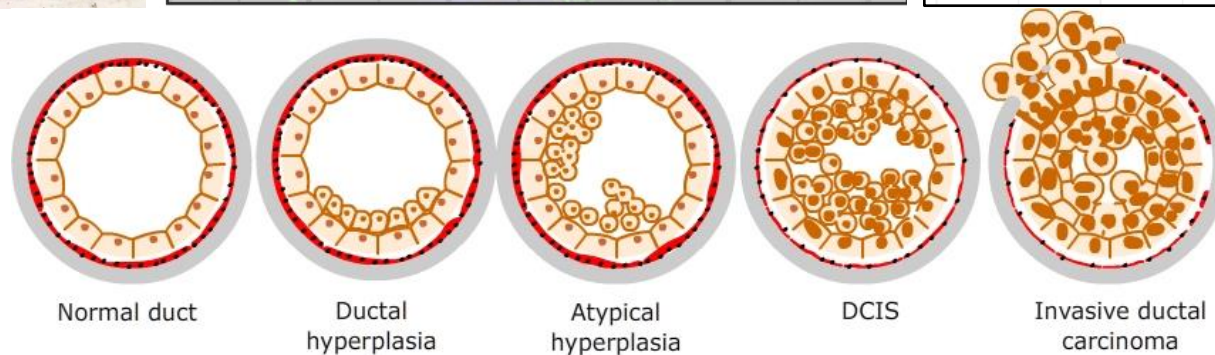
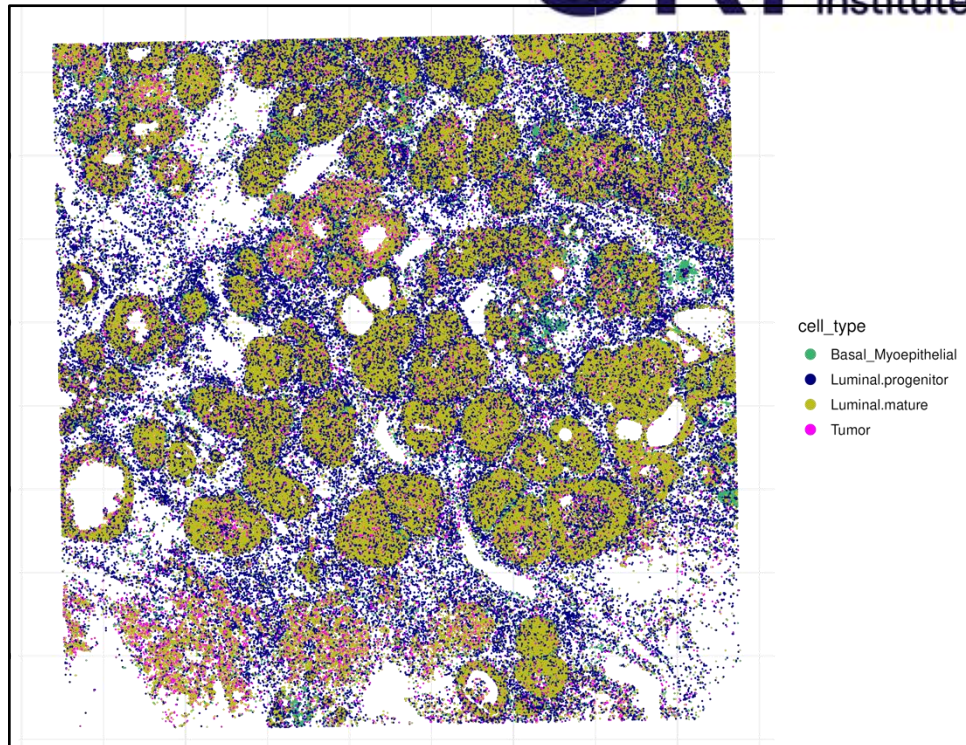
H&E CytAssist Image



Full Object Banksy Clustering



Epithelial Cells only



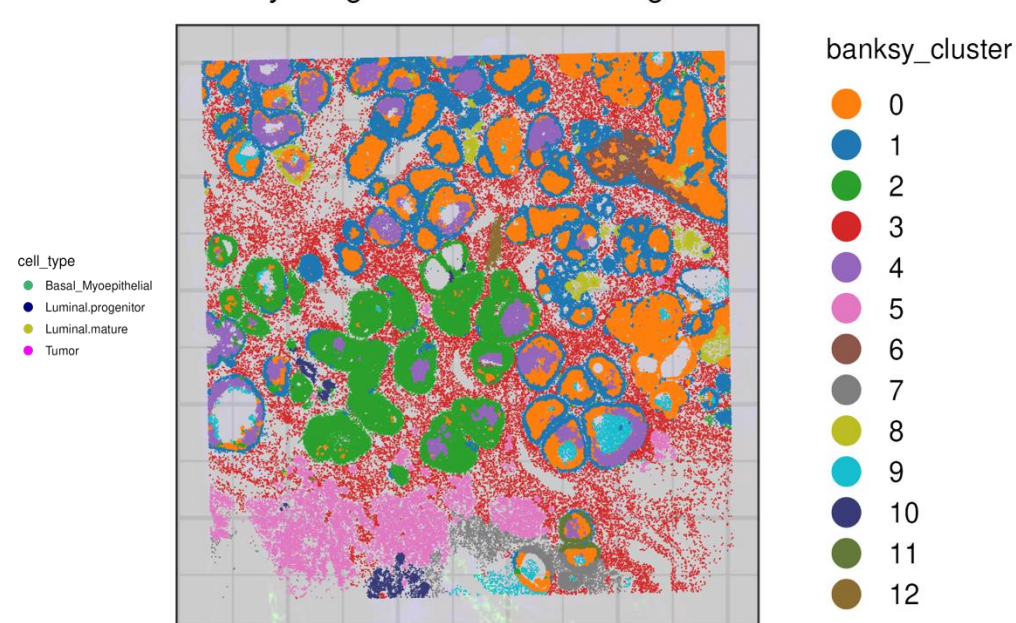
Basement membrane

Myoepithelium

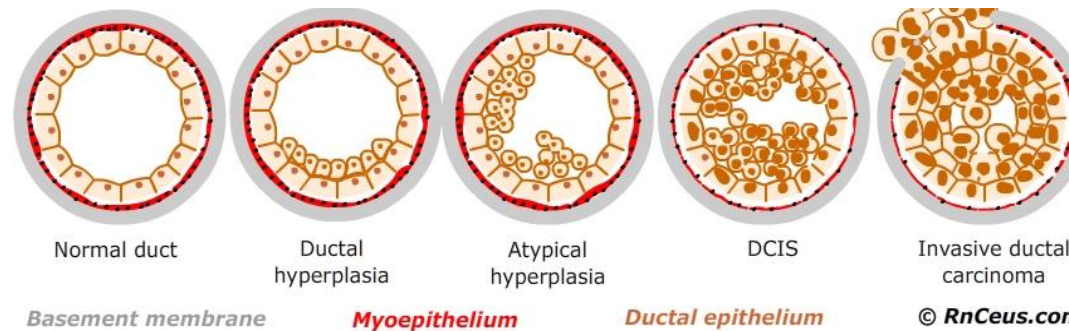
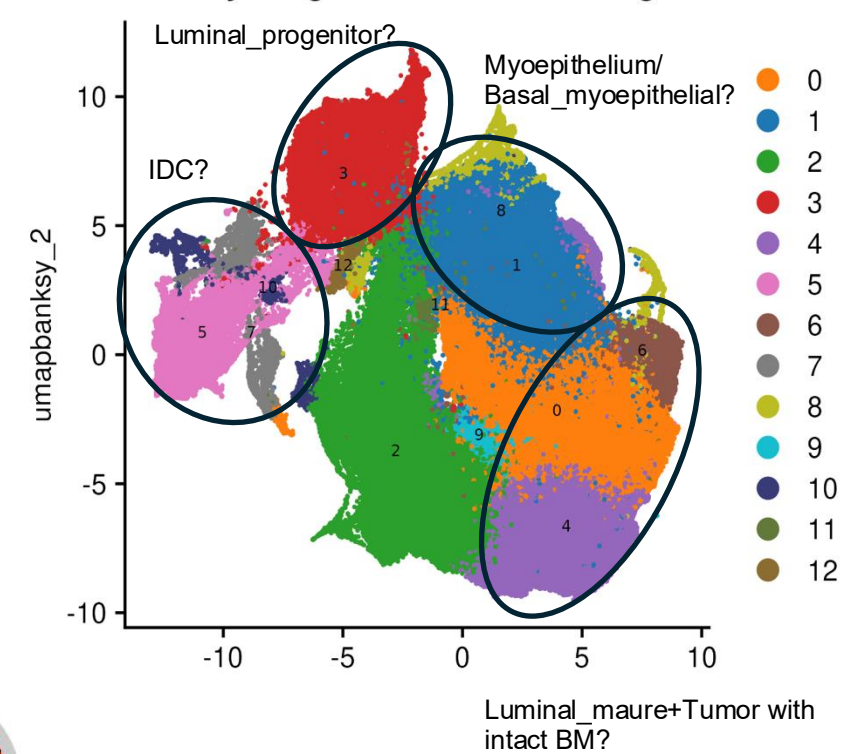
Ductal epithelium

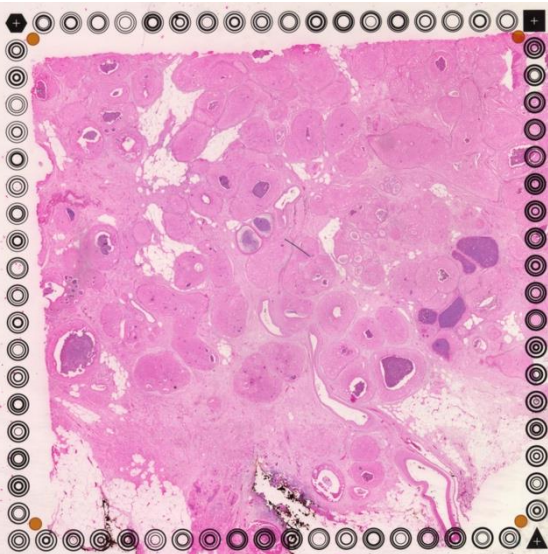
© RnCeus.com

Banksy Neighborhood Clustering

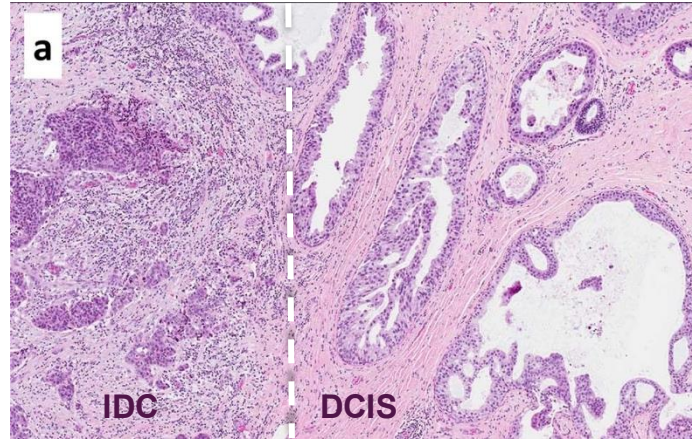


Banksy Neighborhood Clustering



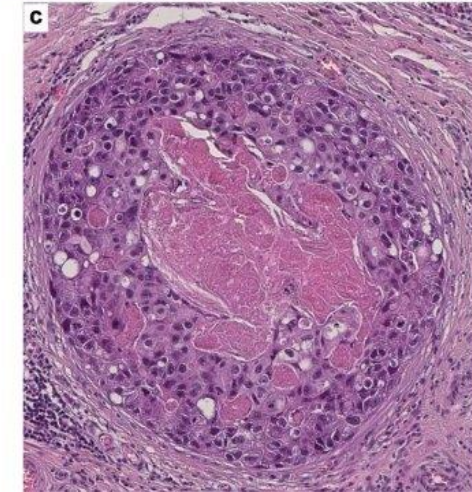
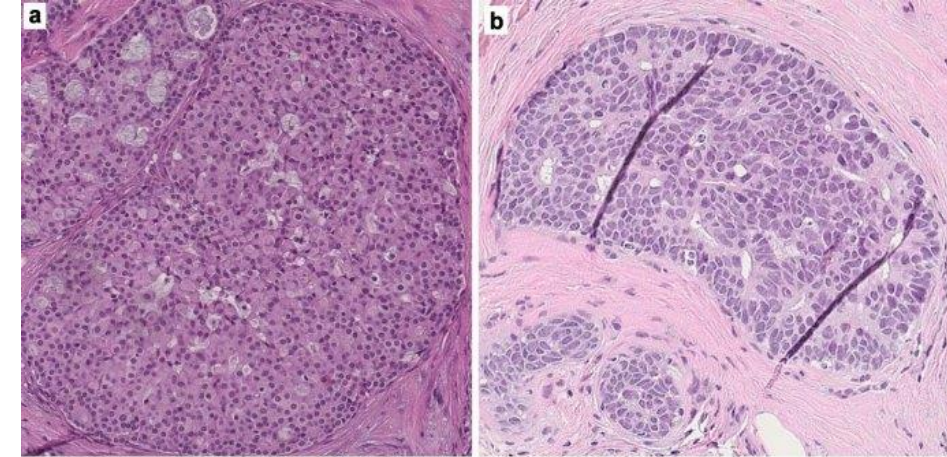


Hanna, W.M., et al. (2019) *Mod Pathol* 32, 896–915



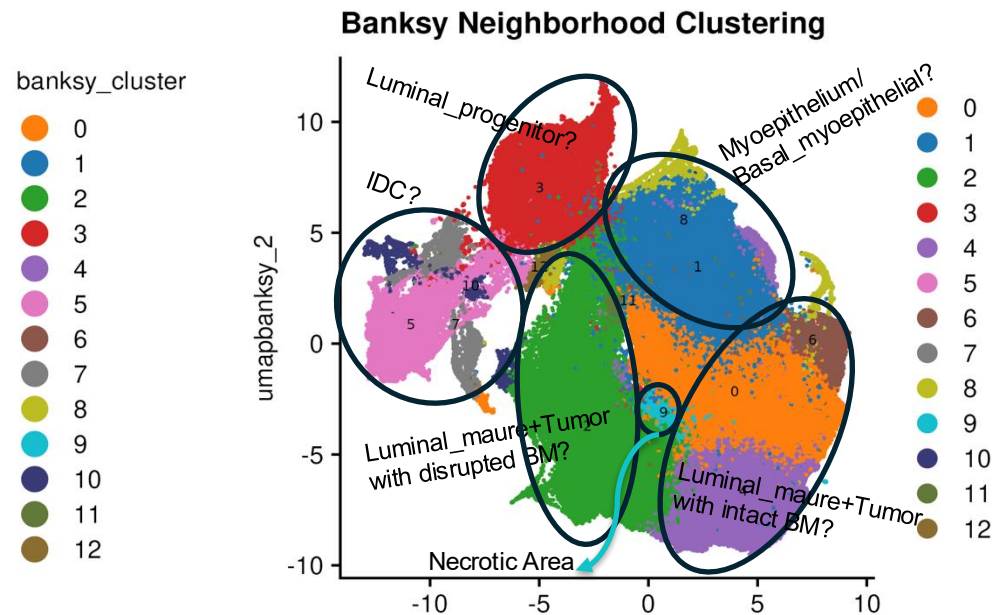
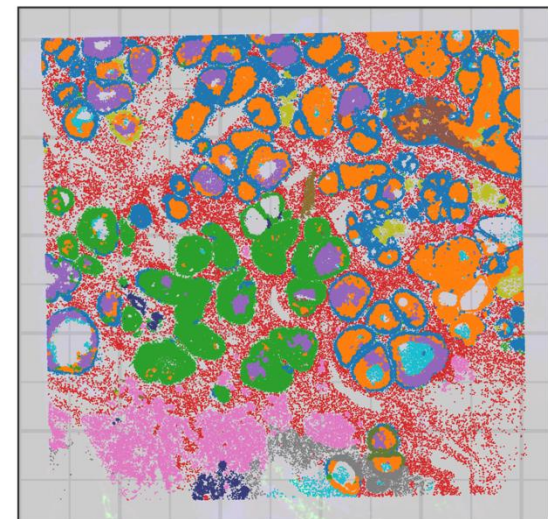
(a) DCIS associated with invasive ductal carcinoma

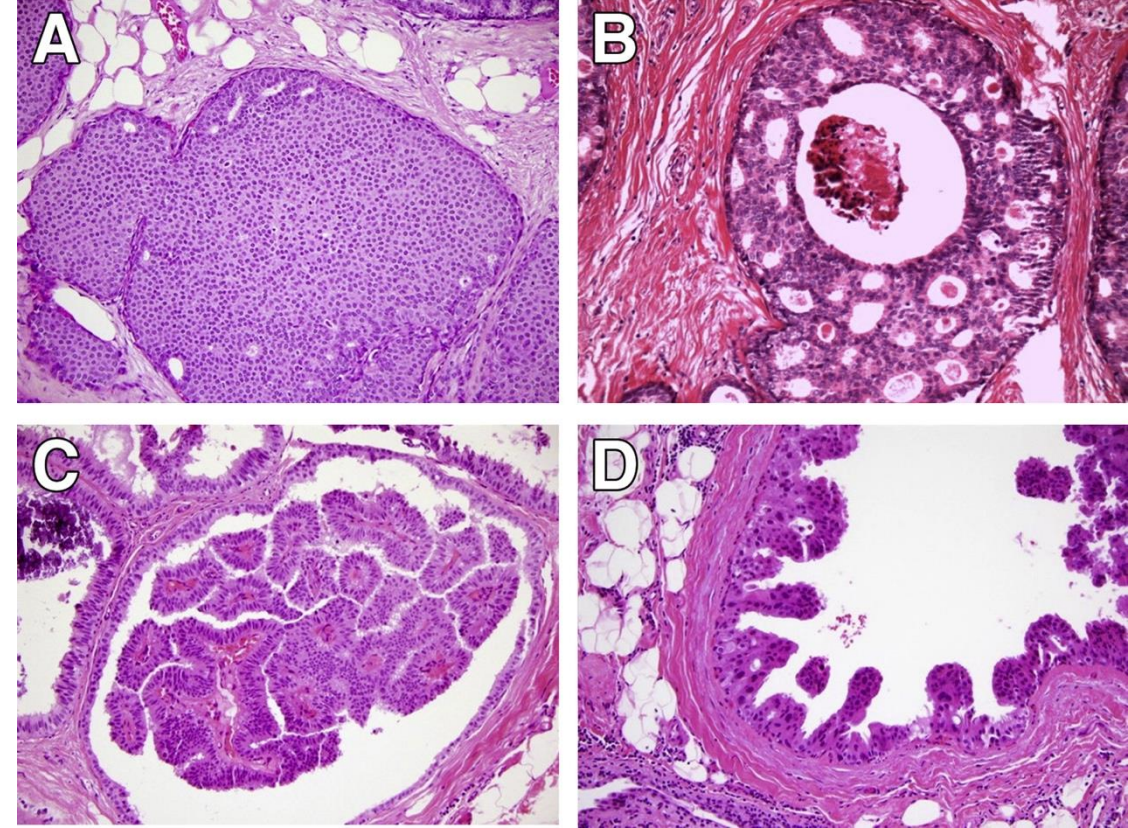
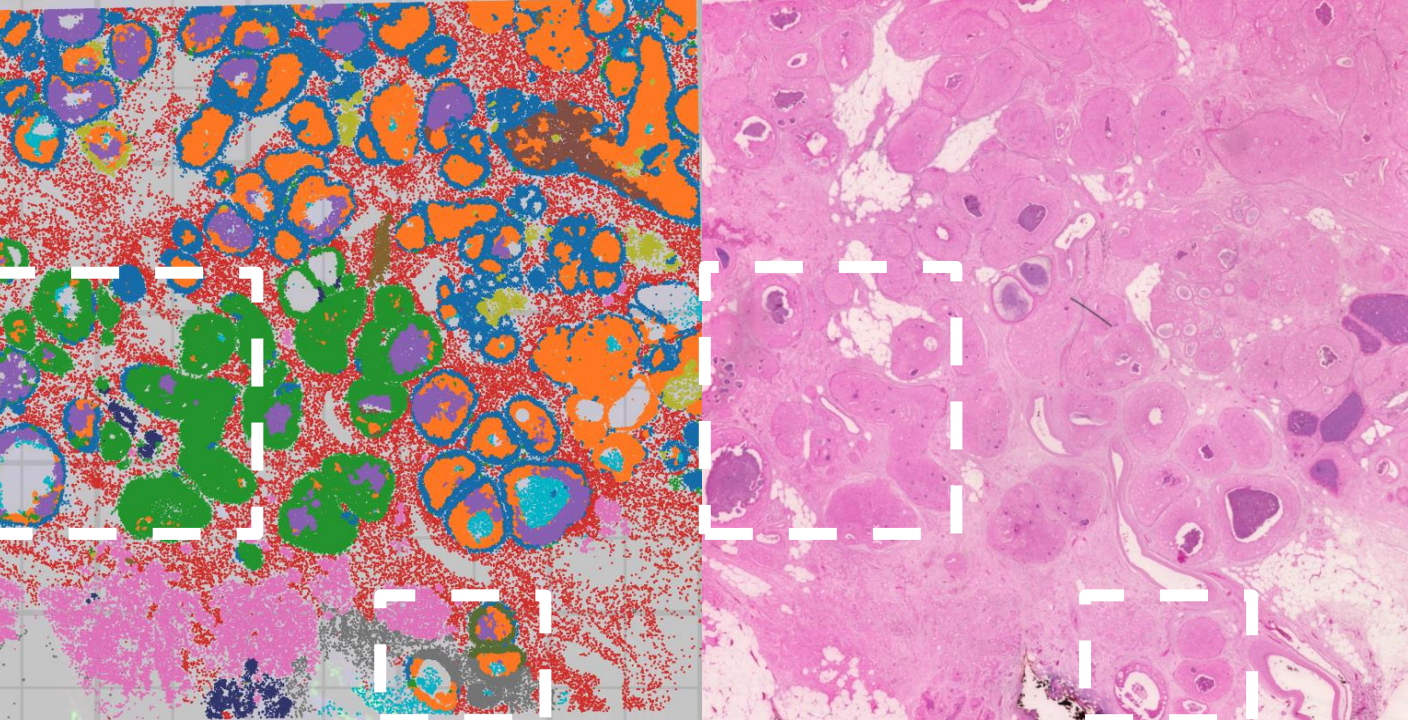
(Castellano, I., Metovic, J. (2018). *Encyclopedia of Pathology*. Springer, Cham.



(a) **Low-grade DCIS** is characterized by small, monomorphic, well-polarized cells, with uniform size. (b) **Intermediate-grade DCIS** consists of cells variable in size, shape, and placement and presents occasional mitotic figures and coarse chromatin. (c) **High-grade DCIS** is composed of highly atypical cells, large in size, with pleomorphic and poorly polarized irregular nuclei and presence of **necrosis** (H&E, 20x)

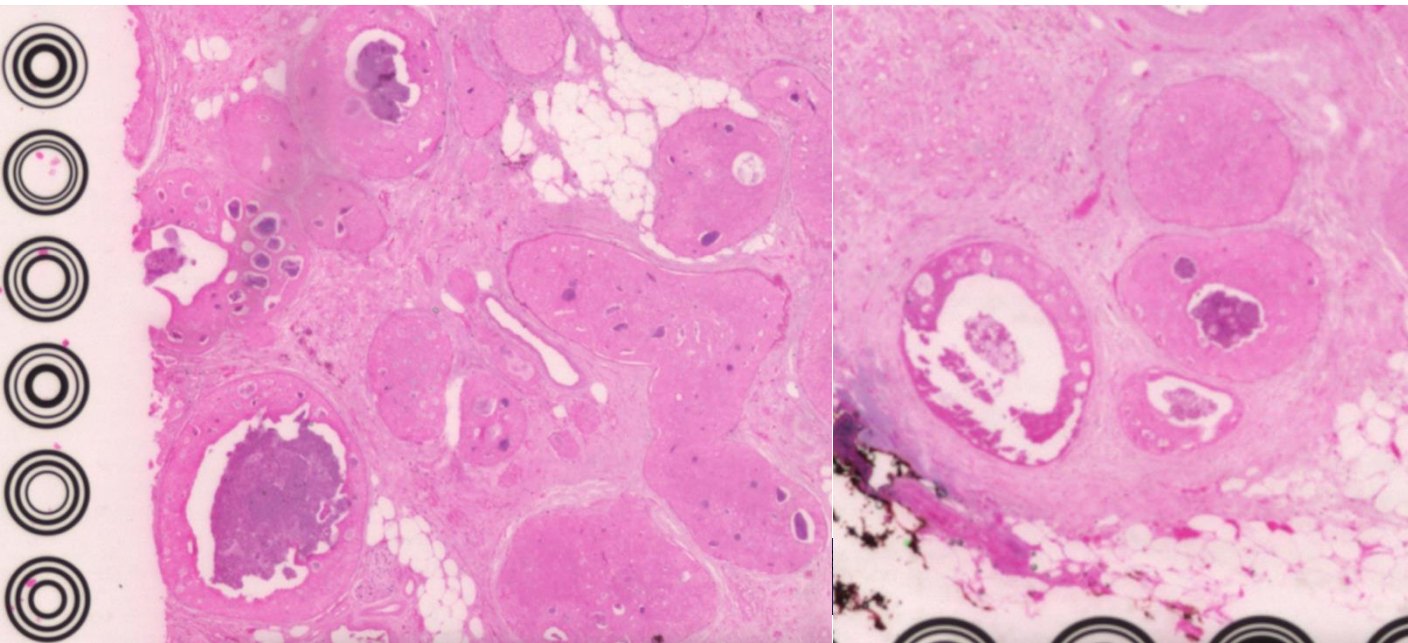
Banksy Neighborhood Clustering





A: Ductal carcinoma *in situ* with solid growth pattern. **B:** Ductal carcinoma *in situ* with cribriform growth pattern and central comedo necrosis. **C:** Ductal carcinoma *in situ* with papillary growth pattern. **D:** Ductal carcinoma *in situ* with micropapillary growth pattern. Hematoxylin and eosin stain was used (**A–D**). Original magnification, $\times 20$

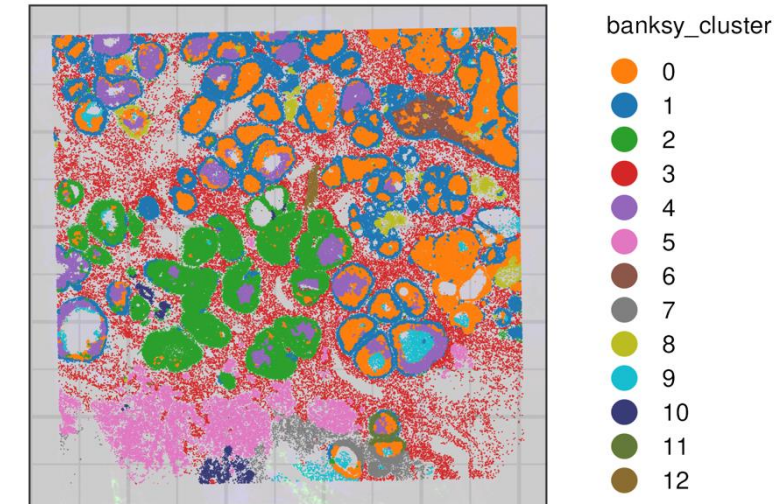
(Sanati S. The American journal of pathology. 2019;189(5):946-55)



some IDC and very few "normal" ducts with a single layer of lumen



Banksy Neighborhood Clustering



Re-assign cell type annotation after AddModuleScore followed by DEG analysis

Importing cluster and cell type annotation to Loupe Browser

```
#####
##### To extract csv files of the cell types and clusters to add to Loupe Browser #####
#####
```

```
metadata <- object@meta.data[, c("seurat_cluster.projected", "banksy_cluster", "cell_type")] %>%
  tibble::rownames_to_column(var = "CellID")
```

```
export_cluster_proj <- metadata[, c("CellID", "seurat_cluster.projected")]
export_cluster_banksy <- metadata[, c("CellID", "banksy_cluster")]
export_celltype <- metadata[, c("CellID", "cell_type")]
```

```
# Write to CSV
```

```
write.csv(export_cluster_proj, file = file.path(sample_dir, paste0(sample_names, "_cluster_proj.csv")), row.names = FALSE)
write.csv(export_cluster_banksy, file = file.path(sample_dir, paste0(sample_names, "_cluster_banksy.csv")), row.names = FALSE)
write.csv(export_celltype, file = file.path(sample_dir, paste0(sample_names, "_cell_type.csv")), row.names = FALSE)
```

Pipeline-generated groups

☐ Graph-Based
 ☐ K-Means

Custom groups

☒ cell_type

☒ All

☒ Basal_Myoepithelial (766)

☒ Luminal.mature (15887)

☒ Luminal.progenitor (8448)

☒ Tumor (1273)



+

Create a new group

Run Differential Expression

Differential Expression Output

Expression Distribution

Feature Table

Heat Map

Up-regulated genes per selected cluster in group **cell_type**, compared between selected cluster(s) themselves.

Feature	Basal_Myoepithelial		Luminal.mature	Luminal.progenit...	Tumor
Name	L2FC	P-Value	L2FC	L2FC	L2FC
DST	4.95	4.06e-23	-1.84	-0.59	-1.25
KRT17	4.78	2.16e-20	-1.66	-0.64	-0.89
KRT14	4.80	1.80e-19	-1.73	-0.53	-1.10

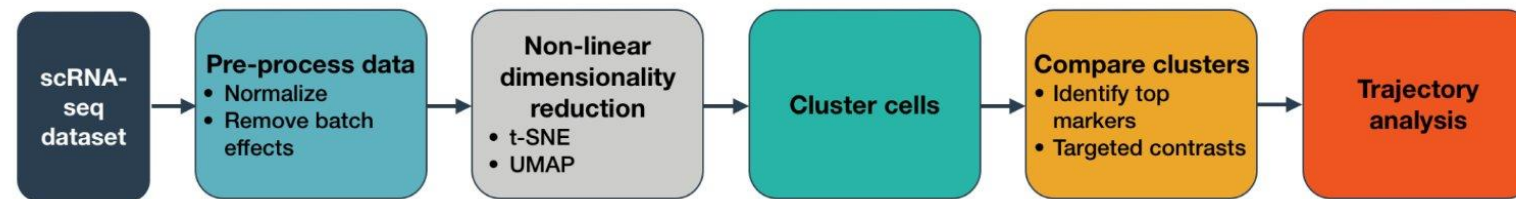
To identify a rare cell type with no prior gene sets in the literature, we can transfer the cluster label to Loupe Browser (or manually select an area annotated by a pathologist) and Run DEG analysis to define a sub-type

Monocle3 Single Cell Trajectory Analysis

[Home](#) / [Docs](#) / [Getting started](#)

Getting started with Monocle 3

Workflow steps at a glance



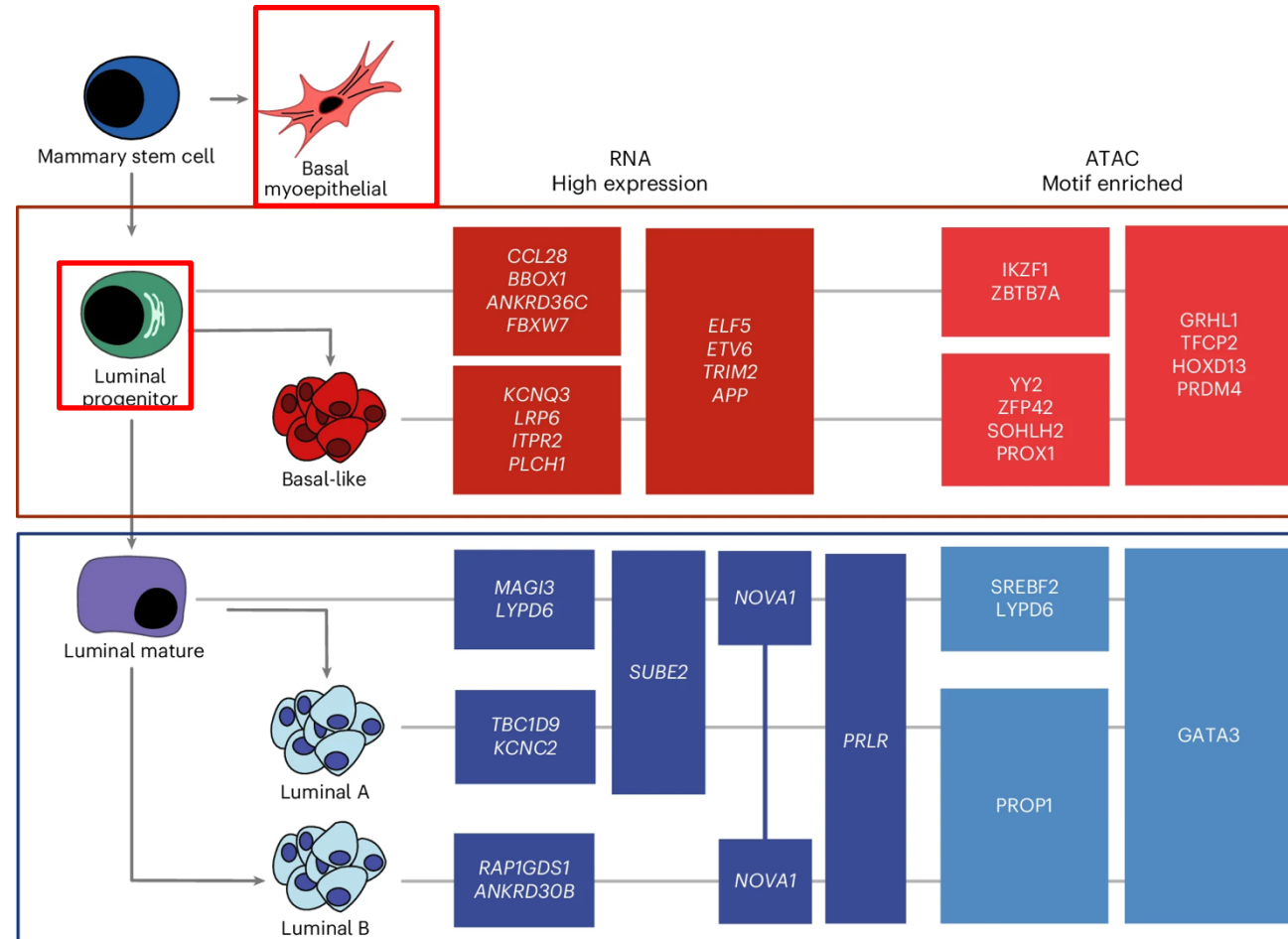
The cell_data_set class

Monocle holds single-cell expression data in objects of the `cell_data_set` class. The class is derived from the Bioconductor `SingleCellExperiment` class, which provides a common interface familiar to those who have analyzed other single-cell experiments with Bioconductor. The class requires three input files:

- `expression_matrix`, a numeric matrix of expression values, where rows are genes, and columns are cells
- `cell_metadata`, a data frame, where rows are cells, and columns are cell attributes (such as cell type, culture condition, day captured, etc.)
- `gene_metadata`, an data frame, where rows are features (e.g. genes), and columns are gene attributes, such as biotype, gc content, etc.

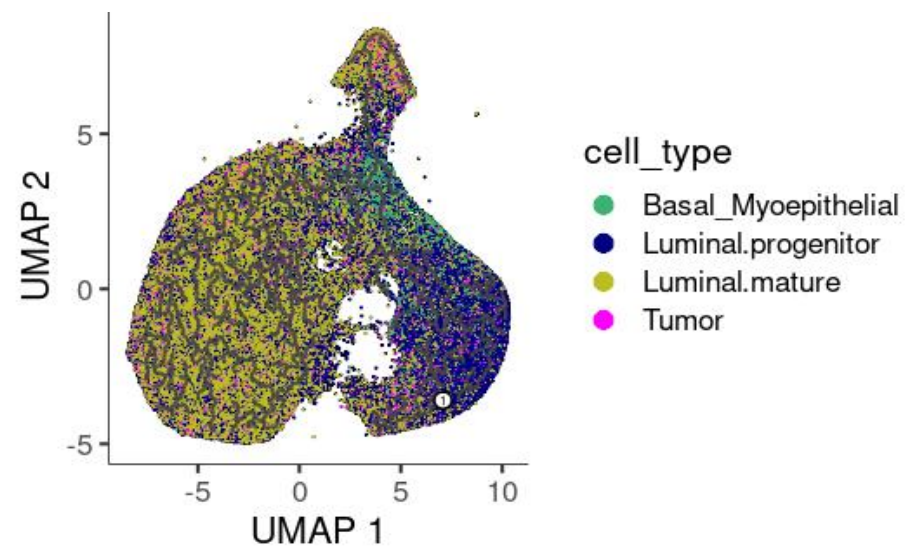
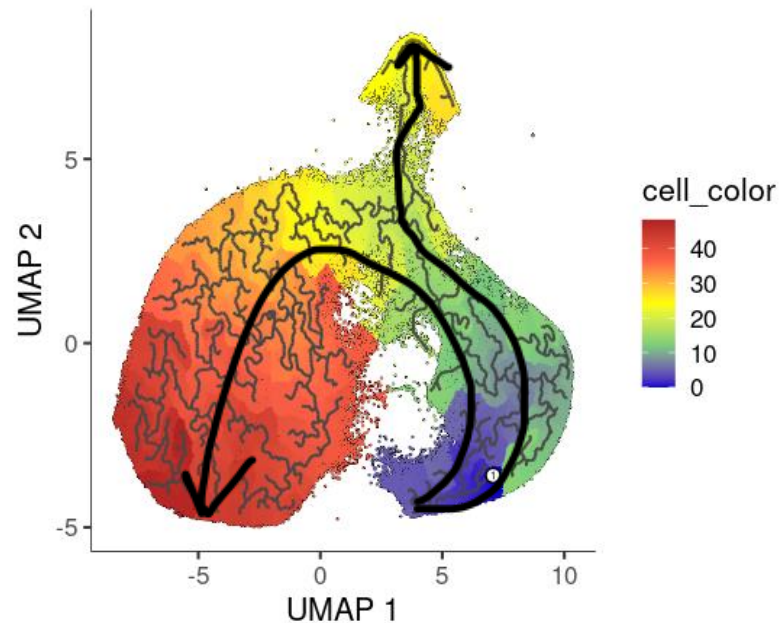
Rather than purifying cells into discrete states experimentally, Monocle uses an algorithm to learn the sequence of gene expression changes each cell must go through as part of a dynamic biological process. Once it has learned the overall "trajectory" of gene expression changes, Monocle can place each cell at its proper position in the trajectory.

If there are multiple outcomes for the process, Monocle will reconstruct a "branched" trajectory. These branches correspond to cellular "decisions", and Monocle provides powerful tools for identifying the genes affected by them and involved in making them.

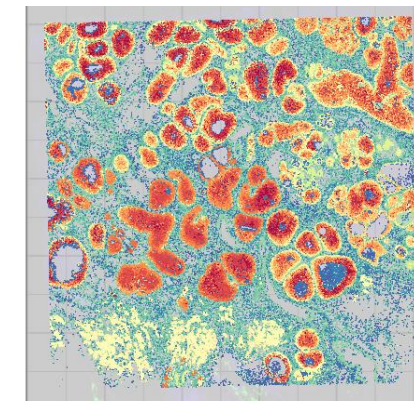
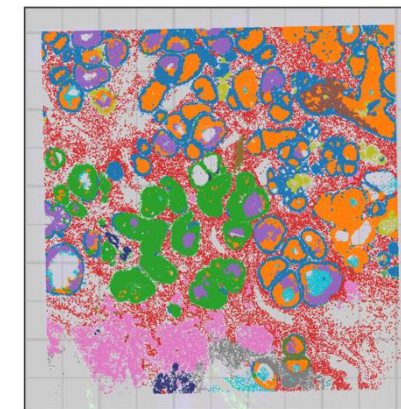


Iglesia MD et al. Nature Cancer. 2024 Nov;5(11):1713-36.

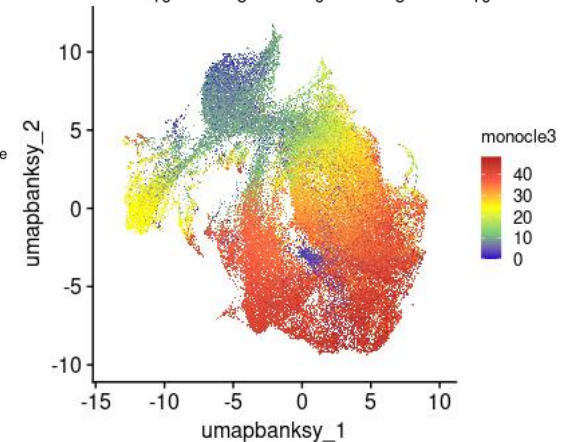
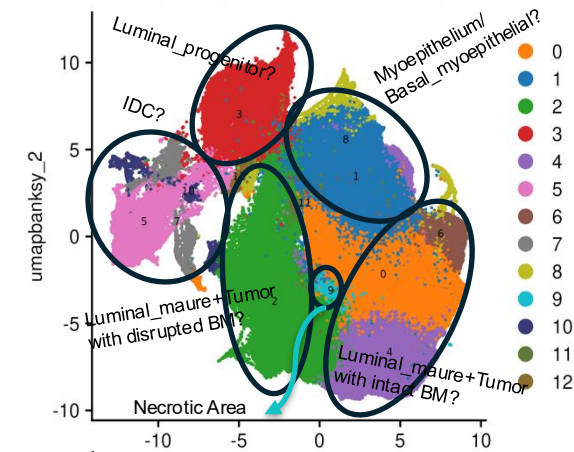
Using both **Luminal.progenitor** and **Basal_Myoepithelial** as the earliest_principal_node



Banksy Neighborhood Clustering



Banksy Neighborhood Clustering



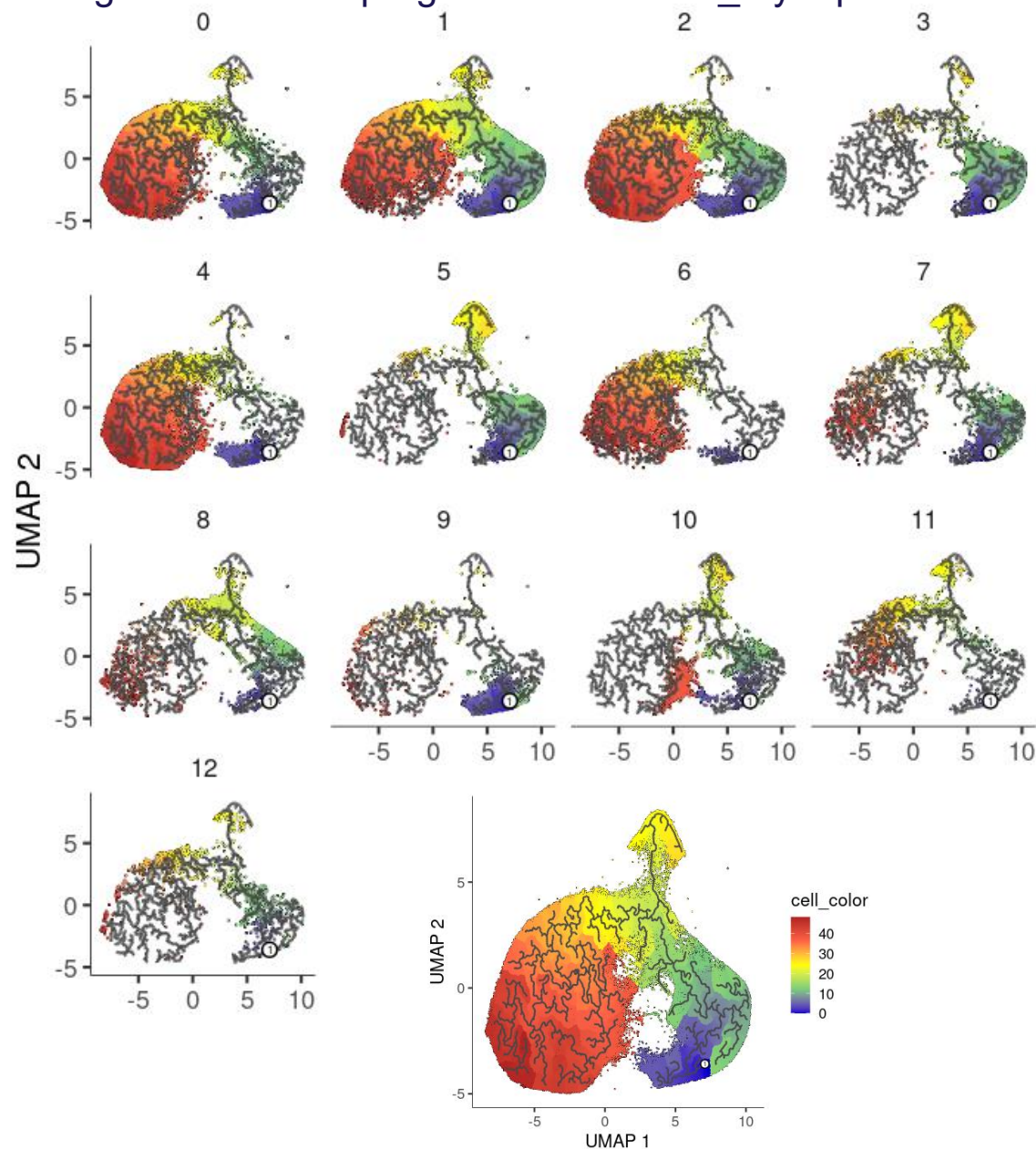
0, 1, 2, 4, 6 seem to be the trajectory from progenitor to well-differentiated DCIS

9 seems to be a poorly-differentiated DCIS

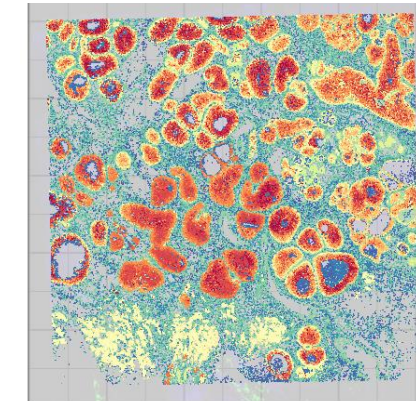
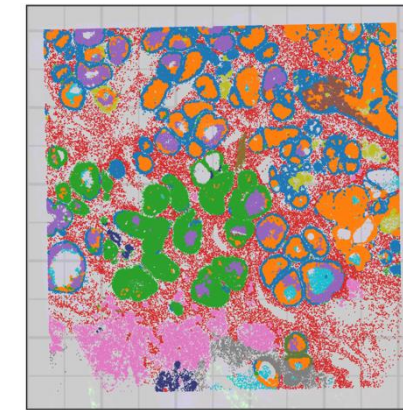
5, 7, 10, 11, 12 seem to be moderately-differentiated IDC?

3 seems to be basal_myoepithelial, and part of **8** seems to be luminal.progenitor (confirm with DEG)

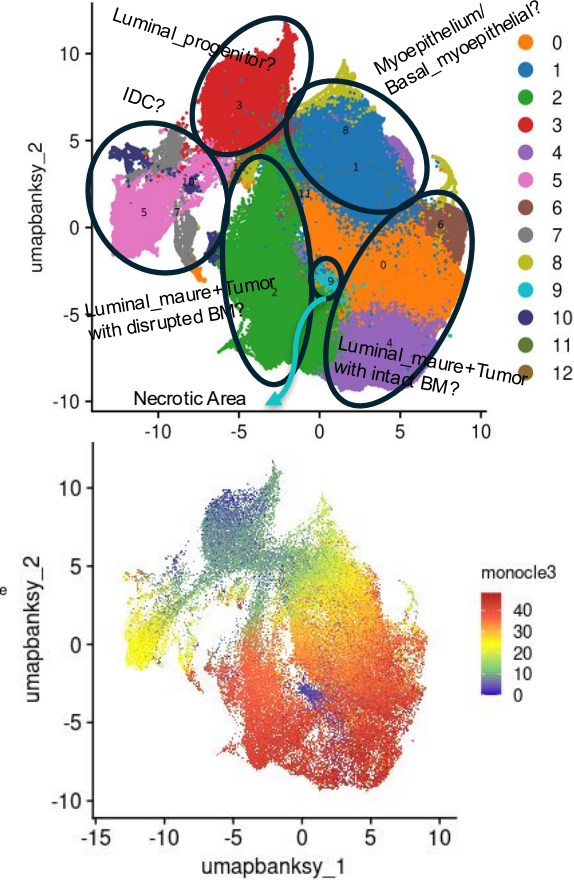
Using both Luminal.progenitor and Basal_Myoepithelial as the earliest_principal_node



Banksy Neighborhood Clustering



Banksy Neighborhood Clustering



0, 1, 2, 4, 6 seem to be the trajectory from progenitor to well-differentiated DCIS

9 seems to be a poorly-differentiated DCIS

5, 7, 10, 11, 12 seem to be moderately-differentiated IDC?

3 seems to be basal_myoepithelial, and part of 8 seems to be luminal.progenitor (confirm with DEG)

Sparsity Issue

- ❖ Consider analyzing the data at 8 μm , 16, and other bin sizes
- ❖ For copy number analysis (for example using InferCNV):
 - Use different aggregate sizes
 - ❑ Can pool 100 bins (10x10 8 μm bins) into 80 μm super bins
 - ❑ Try 16 μm , 40 μm , 80 μm , depending on your data

Hands-on: Unsupervised clustering, cell typing, loupe browser

Summary/Questions

- Sample QC before, during, and after spatial transcriptomics run is very important to ensure good data quality
- Understanding the abundance of the tissue (block/USS/archived slides), the important questions you would like answer (discovery vs validation), plexity needed (whole transcriptome vs custom panel), resolution needed can help decide which tool to use
- Explore the Visium HD data in Loupe Browser, understand the pathology and biology of the tissue, sometimes filtering out the bins with the highest expression might remove important bins (for example tumor areas with WGD, examine each sample carefully before performing the same QC on all samples)
- For Visium HD, merging several samples at 8x8μm bin level can be very computationally intensive. Try to subset to tumor cells/epithelial cells before merging, immune, stroma, etc...
- For CNV analysis, try several aggregate sized (pooling a few adjacent 8x8μm bins, since the data can be very messy at 8x8)

Summary/Questions

Some resources for **Visium HD Spatial Transcriptomics** prior to the session:

- For sample pre-processing, the 10X Genomics [Visium HD Analysis with spaceranger count](#) explains how to run SpaceRanger with automatic and manual alignment. For manual alignment of the Visium HD CytAssist image, please make sure to download [Loupe Browser 8.1.2 \(Nov 18, 2024\)](#).
- To get started with how to perform analysis, visualization, and integration of Visium HD spatial datasets with Seurat, please check this [vignette](#), which also includes information on how to run [Banksy](#) unsupervised neighborhood clustering
- For information on how to construct single-cell trajectories/pseudotime trajectories using Monocle3, please check [this vignette](#)
- For using InferCNV with the Visium HD data, [this vignette](#) illustrates inferring copy number alterations from tumor single cell RNA-Seq data
- For nuclei Segmentation and Custom Binning of Visium HD Gene Expression Data, please refer to [this vignette](#) by 10X. This is the only step that requires working in Python, all the rest of the scripts work with Rstudio. To convert the Seurat object (R) to work with Python (AnnData) please follow [this vignette](#).

Closing and Feedback (15 min)